

claude-windows / 6e9d0958-a2ec-4a8e-bb2d-6cce9833c527

Metadata

- Source: `claude-windows`
- Kind: `claude`
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\6e9d0958-a2ec-4a8e-bb2d-6cce9833c527\subagents\agent-abc6fade3e884711a.jsonl`
- SHA-256: `62356fbc2dd24270b5295e15bef06a0a3681efd41a4e149de821715523387726`
- Source modified: `2026-07-04T00:46:56+00:00`
- Imported at: `2026-07-05T16:48:26+00:00`
- project: `subagents`
- session_id: `6e9d0958-a2ec-4a8e-bb2d-6cce9833c527`

Transcript

1. user (2026-07-04T00:45:04.045Z)

Code-review finder. Target: PR #458 of badminton-highlight-indexer. Diff: `cd C:/Users/avidu/Projects/khelsutra-guru/badminton-highlight-indexer && gh pr diff 458`. PR branch = `origin/codex/a7-a9-quality-chain`. Files at scratchpad/pr458_fusion_golden.py, pr458_fusion_audio.py.

ANGLE ? altitude / right-depth. Context: A7 = regenerate full-fusion feature JSONs for the 9 newer golden videos; A8 = rerun fusion/served decisions; A9 = Gen-0 shadow training. This PR is the eval TOOLING to run that.

1. The full-fusion vs windowing-only distinction is detected at RUN time by re-reading each JSON (`is\_full\_fusion\_feature\_file`). Is a runtime schema-sniff the right depth, or should the feature JSONs carry an explicit schema/version tag written at generation time (so consumers don't guess)? The doc's own §9 Q5 asked exactly this ("explicit capability tags"). Does this PR resolve or paper over that open question?

2. `attach\_source\_video\_paths` introduces a SECOND source of truth for video paths (the collector source manifest) layered on top of A2's golden_manifest path resolver. Is stacking a second path-attach mechanism the right depth, or should the manifest resolver (golden_manifest.py) own source-video resolution too?

3. `--max-gpu-temp-c` thermal abort is wired through extract_video's telemetry _progress. Is thermal safety at the right layer (per-eval-tool), or should it live in the shared detector/telemetry layer so ALL GPU passes get it, not just fusion_golden?

4. Tracker (docs/ALPHA_LAUNCH_READINESS.md): A6/A7/A8/A9 all flipped to states in "this PR" but the PR is a DRAFT and A9 explicitly leaves TemporalMaxer unimplemented with an open scope question ? does marking A9 "IN PROGRESS | this PR" while punting the core deliverable misrepresent status per the tracker's own §7 conventions?

Only flag concrete altitude problems naming the deeper alternative. Return up to 6 candidates as JSON [{file, line, summary, failure_scenario}]. [] if none. Final message = only the JSON array.

2. assistant (2026-07-04T00:45:05.877Z)

I'll analyze this PR for altitude/right-depth issues. Let me start by fetching the PR diff and examining the files.

3. user (2026-07-04T00:45:07.884Z)

<persisted-output>

Output too large (51.9KB). Full output saved to: C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\6e9d0958-a2ec-4a8e-bb2d-6cce9833c527\tool-results\bzsjson4r.txt

Preview (first 2KB):

```
diff --git a/SESSION_HANDOFF.md b/SESSION_HANDOFF.md
index 31ea4c14..a50cc8a1 100644
--- a/SESSION_HANDOFF.md
+++ b/SESSION_HANDOFF.md
@@ -2,49 +2,51 @@
```

You are here

-A6 is implemented and open for review as PR #455:
-`https://github.com/Khelsutra/badminton-highlight-indexer/pull/455`.
+A6 PR #455 is merged. This checkout is on `codex/a7-a9-quality-chain`, fast-forwarded to
+`origin/master` at `c5fc27c`, with tracked edits in:

-The branch is `codex/a6-ablation-cli`, rebased onto `origin/master` after A5
-merged as `c8eec7c` / PR #454. It fixes the `python -m backend.eval.ablation`
-circular import by making `ablation.export\_pdf` a lazy wrapper, adds a
-module-entrypoint regression test, and records the n=15 R2/tolF1 contrast in
-`docs/ALPHA\_LAUNCH\_READINESS.md`.
+- `backend/eval/fusion\_golden.py`
+- `backend/eval/fusion\_audio.py`
+- `tests/test\_fusion\_compare.py`
+- `tests/test\_audio\_features.py`
+- `docs/ALPHA\_LAUNCH\_READINESS.md`
+- `SESSION\_HANDOFF.md`

-Validation completed locally on 2026-07-03:
-`ruff check .`, `python -m mypy backend training`, and
-`python run\_all\_tests.py --cov=backend --cov=training --cov-report=term --cov-fail-under=70`
-all passed. Full suite result: 1609 passed, 6 skipped, 85.65% coverage.
+A7/A8/A9 have been run locally against the 15-video corpus, and draft PR #458 is open:
+`https://github.com/Khelsutra/badminton-highlight-indexer/pull/458`. The generated
+feature/model artifacts live under ignored `output/` and `artifacts/` paths and should not
+be committed.

Next steps / open threads

-1. Review/merge PR #455 for A6.
-2. A7 is the next indexer-owned quality item: regenerate or upconvert
- full-fusion feature JSONs for the newer nine golden videos so fusion/person
- /audio analyses can run on all 15 videos. GPU is approved on this machine.
-3. A8 follows A7: rerun `fusion\_compare`, group ablation, and served contrast;
- explicitly promote or reject fusion/person/Gate-A served-default decisions.
...
</persisted-output>

4. assistant (2026-07-04T00:45:10.220Z)

Let me read the full diff and the specific files mentioned.

5. user (2026-07-04T00:45:11.801Z)

```
1      diff --git a/SESSION_HANDOFF.md b/SESSION_HANDOFF.md
2      index 31ea4c14..a50cc8a1 100644
3      --- a/SESSION_HANDOFF.md
4      +++ b/SESSION_HANDOFF.md
5      @@ -2,49 +2,51 @@
6
7      ## You are here
8
9      -A6 is implemented and open for review as PR #455:
10     -`https://github.com/Khelsutra/badminton-highlight-indexer/pull/455`.
11     +A6 PR #455 is merged. This checkout is on `codex/a7-a9-quality-chain`, fast-forwarded
12     to
13     +`origin/master` at `c5fc27c`, with tracked edits in:
14
15     -The branch is `codex/a6-ablation-cli`, rebased onto `origin/master` after A5
16     -merged as `c8eec7c` / PR #454. It fixes the `python -m backend.eval.ablation`
17     -circular import by making `ablation.export_pdf` a lazy wrapper, adds a
18     -module-entrypoint regression test, and records the n=15 R2/tolF1 contrast in
19     -`docs/ALPHA_LAUNCH_READINESS.md`.
20     +- `backend/eval/fusion_golden.py`
```

```

20 +- `backend/eval/fusion_audio.py`
21 +- `tests/test_fusion_compare.py`
22 +- `tests/test_audio_features.py`
23 +- `docs/ALPHA_LAUNCH_READINESS.md`
24 +- `SESSION_HANDOFF.md`
25
26 -Validation completed locally on 2026-07-03:
27 -`ruff check .`, `python -m mypy backend training`, and
28 -`python run_all_tests.py --cov=backend --cov=training --cov-report=term --cov-fail-
under=70`
29 -all passed. Full suite result: 1609 passed, 6 skipped, 85.65% coverage.
30 +A7/A8/A9 have been run locally against the 15-video corpus, and draft PR #458 is open:
31 +`https://github.com/Khelsutra/badminton-highlight-indexer/pull/458`. The generated
32 +feature/model artifacts live under ignored `output/` and `artifacts/` paths and should
not
33 +be committed.
34
35 ## Next steps / open threads
36
37 -1. Review/merge PR #455 for A6.
38 -2. A7 is the next indexer-owned quality item: regenerate or upconvert
39 -   full-fusion feature JSONs for the newer nine golden videos so fusion/person
40 -   /audio analyses can run on all 15 videos. GPU is approved on this machine.
41 -3. A8 follows A7: rerun `fusion_compare`, group ablation, and served contrast;
42 -   explicitly promote or reject fusion/person/Gate-A served-default decisions.
43 -4. A3 is owned by another agent; do not duplicate that collector/vault work.
44 +1. Address review comments on draft PR #458; convert to ready only after owner/reviewer
agrees
45 +   the A9 TemporalMaxer scope question is settled.
46 +2. Decide whether A9 is considered complete with the current Gen-0 non-serving shadow
47 +   bundle or whether TemporalMaxer needs a follow-up scorer-swap PR.
48 +3. Do not touch A1; another agent owns it.
49 +4. A3 is in collector PR #62 and records all 15 source videos; the two Boxhill
originals
50 +   were verified on `F:\`.
51
52 ## Ramp-up kit
53
54 - `docs/ALPHA_LAUNCH_READINESS.md`
55 - `docs/data_pipeline/GOLDEN_VIDEOS.md`
56 - `docs/R2_EVAL_METRIC_DESIGN.md`
57 - `backend/eval/ablation.py`
58 - `backend/eval/ablation_pdf.py`
59 - `backend/eval/rally_seg_eval.py`
60 - `tests/test_ablation.py`
61 - `output/human_lovo_manifest.json`
62 + `backend/eval/fusion_golden.py`
63 + `backend/eval/fusion_audio.py`
64 + `training/gen0/harness.py`
65 + `eval_baselines/heuristic_lovo_n15.json`
66 + `output/a7_remaining_default.out.log`
67 + `output/a8_audio_augment.out.log`
68 + `output/a6_r2_ablation.json`
69 + `artifacts/rally_tal_gen0/0.1.1-n15-shadow.20260704/manifest.json`
70
71 ## Key decisions
72
73 - R2/tolF1 is supported for the baseline-off to served milestone contrast on
74 -   the 15-video corpus: `Delta tolF1 = +0.225`, CI `[+0.165, +0.283]`,
75 -   sign `14/0`, `p = 0.000`.
76 - This does not approve fusion/person/audio default changes. Current full-fusion
77 -   ablation still runs on only six videos because the newer nine do not yet have
78 -   full-fusion feature schema.
79 - Branching discipline remains important: always fetch/pull first, branch from
80 -   `origin/master`, stage explicit paths, and verify `origin/master..HEAD` before

```

```

81     - pushing.
82     +- A7: full-fusion schema and audio features are present locally for all 15 videos.
83     +- A8: person fusion and audio are suggestive but not statistically clean; do not
promote
84     + either as a default. Gate A regresses under served windowing; keep it opt-
in/analysis-only.
85     +- R2/cap/inactivity remains the strongest quality result on n15: F1 `0.237 -> 0.472`,
86     + tolF1 `0.138 -> 0.364`, merge rate `0.638 -> 0.135`; net over-seg guard passes.
87     +- A9: Gen-0 LogReg/TAL n15 shadow bundle beats the heuristic mean (`0.551 > 0.4456`)
88     + but is non-serving (`9/15` paired wins, `p=0.3036`; #172 and WASB C3 still block).
89     +- Branching discipline remains important: fetch/pull first, stage explicit paths, and
90     + verify `origin/master..HEAD` contains only this branch's commits before pushing.
91     diff --git a/backend/eval/fusion_audio.py b/backend/eval/fusion_audio.py
92     index b5c65ec4..7dff0c94 100644
93     --- a/backend/eval/fusion_audio.py
94     +++ b/backend/eval/fusion_audio.py
95     @@ -17,14 +17,17 @@
96         import json
97         import logging
98         import os
99     -from typing import Any, Dict, List
100     +from typing import Any, Dict, List, Optional, Sequence
101
102     from backend.eval import experiment
103     from backend.eval.fusion_compare import delta_stats
104     from backend.eval.fusion_golden import (
105         PERSON_FEATURE_NAMES,
106         SHUTTLE_FEATURE_NAMES,
107     +     SOURCE_MANIFEST_ENV,
108     +     _derive_video_path,
109     +     _filter_specs,
110     +     attach_source_video_paths,
111     )
112     from backend.eval.golden_manifest import load_resolved_manifest
113     from backend.pipeline.audio.hit_detector import detect_hits_in_video
114     @@ -33,9 +36,24 @@
115     logger = logging.getLogger(__name__)
116
117
118     -def augment(manifest_path: str, features_dir: str, k: float = 2.5) -> List[str]:
119     +def augment(
120     +     manifest_path: str,
121     +     features_dir: str,
122     +     k: float = 2.5,
123     +     source_manifest: Optional[str] = None,
124     +     only: Sequence[str] = (),
125     +) -> List[str]:
126         """Add the 3 audio features to each existing ``<name>_golden_features.json``
window."""
127         specs = load_resolved_manifest(manifest_path)
128     +     source_manifest = source_manifest or os.environ.get(SOURCE_MANIFEST_ENV)
129     +     if source_manifest:
130     +         updated = attach_source_video_paths(specs, source_manifest)
131     +         print(
132     +             f"[fusion_audio] attached {updated}/{len(specs)} source video path(s) "
133     +             f"from {source_manifest}",
134     +             flush=True,
135     +         )
136     +     specs = _filter_specs(specs, only)
137     +     written: List[str] = []
138     +     for spec in specs:
139     +         jp = os.path.join(features_dir, f"{spec['name']}_golden_features.json")
140     @@ -45,6 +63,10 @@ def augment(manifest_path: str, features_dir: str, k: float = 2.5) ->
List[str]:
141         )

```

```

142         continue
143         video = _derive_video_path(spec)
144         +         if not os.path.isfile(video):
145         +             raise FileNotFoundError(
146         +                 f"{spec.get('name')}: source video not found for audio augmentation:
{video}"
147         +             )
148         print(
149             f"[fusion_audio] {spec['name']}: detecting audio hits (k={k})...",
150             flush=True,
151         @@ -129,6 +151,21 @@ def main() -> None:
152             ap.add_argument("--manifest", default="output/human_lovo_manifest.json")
153             ap.add_argument("--features-dir", default="output")
154             ap.add_argument("--k", type=float, default=2.5, help="hit-detector sensitivity")
155         +         ap.add_argument(
156         +             "--source-manifest",
157         +             default=None,
158         +             help=(
159         +                 "collector golden_source_videos.json; overrides video paths for
source/proxy "
160         +                 f"videos (or set {SOURCE_MANIFEST_ENV})"
161         +             ),
162         +         )
163         +         ap.add_argument(
164         +             "--only",
165         +             action="append",
166         +             default=[],
167         +             metavar="VIDEO_ID",
168         +             help="augment one manifest video by name; repeatable",
169         +         )
170         ap.add_argument(
171             "--augment",
172             action="store_true",
173         @@ -142,7 +179,13 @@ def main() -> None:
174             args = ap.parse_args()
175
176             if args.augment:
177                 -             augment(args.manifest, args.features_dir, args.k)
178                 +             augment(
179                 +                 args.manifest,
180                 +                 args.features_dir,
181                 +                 args.k,
182                 +                 source_manifest=args.source_manifest,
183                 +                 only=args.only,
184                 +             )
185             if args.compare:
186                 videos = load_videos_with_audio(args.features_dir)
187                 if len(videos) < 2:
188                 diff --git a/backend/eval/fusion_golden.py b/backend/eval/fusion_golden.py
189                 index aee6d7e5..0666a0ca 100644
190                 --- a/backend/eval/fusion_golden.py
191                 +++ b/backend/eval/fusion_golden.py
192                 @@ -21,7 +21,7 @@
193                 import json
194                 import os
195                 import time
196                 -from typing import Any, Dict, List, Optional
197                 +from typing import Any, Dict, List, Optional, Sequence
198
199                 from backend.eval import distill_local
200                 from backend.eval.calibrate_wasb import load_golden_gts
201                 @@ -36,6 +36,100 @@
202                 SHUTTLE_FEATURE_NAMES = list(distill_local.FEATURE_NAMES)
203                 # The 5 person-cue features (fusion_features.PERSON_FEATURE_NAMES).
204                 PERSON_FEATURE_NAMES = list(ff.PERSON_FEATURE_NAMES)

```

```

205 +SOURCE_MANIFEST_ENV = "RALLY_GOLDEN_SOURCE_MANIFEST"
206 +
207 +
208 +def is_full_fusion_feature_file(path: str) -> bool:
209 +     """True when an existing golden feature JSON already has shuttle+person cue dicts.
210 +
211 +     Corpus-growth runs can leave leaner windowing-only JSONs with the same filename
pattern.
212 +     Those are useful for gts/windowing analysis, but they are not the replayable fusion
213 +     substrate that ``fusion_compare`` and ``ablation`` need.
214 +     """
215 +
216 +     try:
217 +         with open(path, encoding="utf-8") as f:
218 +             data = json.load(f)
219 +     except (OSError, json.JSONDecodeError, TypeError):
220 +         return False
221 +     candidates = data.get("candidates")
222 +     if not isinstance(candidates, list):
223 +         return False
224 +     if not candidates:
225 +         return True
226 +     return all(
227 +         isinstance(candidate, dict)
228 +         and "shuttle" in candidate
229 +         and "person" in candidate
230 +         for candidate in candidates
231 +     )
232 +
233 +
234 +def load_source_video_paths(source_manifest_path: Optional[str]) -> Dict[str, str]:
235 +     """Load ``video_id -> local_path`` from the collector golden-source manifest.
236 +
237 +     The collector records source/proxy ownership and full-file MD5s. Only entries
marked
238 +     ``present_local`` with a concrete ``local_path`` can drive local GPU extraction.
239 +     """
240 +
241 +     if not source_manifest_path:
242 +         return {}
243 +     with open(source_manifest_path, encoding="utf-8") as f:
244 +         data = json.load(f)
245 +         entries = data.get("videos", data) if isinstance(data, dict) else data
246 +         if not isinstance(entries, list):
247 +             raise ValueError(f"source manifest must contain a videos list:
{source_manifest_path}")
248 +
249 +         paths: Dict[str, str] = {}
250 +         for entry in entries:
251 +             if not isinstance(entry, dict):
252 +                 continue
253 +             video_id = entry.get("video_id") or entry.get("name")
254 +             local_path = entry.get("local_path")
255 +             if not video_id or not local_path:
256 +                 continue
257 +             if entry.get("source_status", "present_local") != "present_local":
258 +                 continue
259 +             paths[str(video_id)] = str(local_path)
260 +         return paths
261 +
262 +
263 +def attach_source_video_paths(
264 +     specs: Sequence[Dict[str, Any]], source_manifest_path: Optional[str]
265 +) -> int:
266 +     """Attach explicit ``video_path`` fields to manifest specs from the source

```

```

manifest.
267 +
268 + Returns the number of specs updated. The collector source manifest is authoritative
269 + for local source/proxy video location and overrides derived/stale manifest paths.
270 + """
271 +
272 + source_paths = load_source_video_paths(source_manifest_path)
273 + updated = 0
274 + for spec in specs:
275 +     name = str(spec.get("name", ""))
276 +     source_path = source_paths.get(name)
277 +     if not source_path:
278 +         continue
279 +     if spec.get("video_path") == source_path:
280 +         continue
281 +     spec["video_path"] = source_path
282 +     updated += 1
283 + return updated
284 +
285 +
286 +def _filter_specs(
287 +    specs: Sequence[Dict[str, Any]], only: Sequence[str]
288 +) -> List[Dict[str, Any]]:
289 +    if not only:
290 +        return list(specs)
291 +    wanted = list(dict.fromkeys(only))
292 +    wanted_set = set(wanted)
293 +    filtered = [spec for spec in specs if str(spec.get("name")) in wanted_set]
294 +    found = {str(spec.get("name")) for spec in filtered}
295 +    missing = [name for name in wanted if name not in found]
296 +    if missing:
297 +        raise ValueError(f"--only requested unknown video(s): {' ', '.join(missing)}")
298 +    return filtered
299
300
301     def _derive_video_path(spec: Dict[str, Any]) -> str:
302 @@ -58,6 +152,9 @@ def extract_video(
303         iou: float = 0.5,
304         log_every_sec: float = 15.0,
305         telemetry: Optional[Any] = None,
306         use_gpu: bool = True,
307         detector_model: Optional[str] = None,
308         max_gpu_temp_c: Optional[float] = None,
309     ) -> Dict[str, Any]:
310         """One golden video -> a feature record (shuttle + person features per candidate
window).
311
312 @@ -77,11 +174,18 @@ def extract_video(
313         # (parity with the shuttle net-crossing gate) rather than hard-coding an
orientation.
314         axis, net_px, _span = rally_gate.estimate_play_axis(points)
315         video_path = _derive_video_path(spec)
316         if not os.path.isfile(video_path):
317             raise FileNotFoundError(
318                 f"{spec.get('name')}: source video not found for fusion extraction:
{video_path}"
319         +
320         )
321         if detector is None:
322             from backend.pipeline.detectors.person_detector import PersonDetector
323
324         - detector = PersonDetector({"indexing": {"use_gpu": True}})
325         + indexing_cfg: Dict[str, Any] = {"use_gpu": use_gpu}
326         + if detector_model:
327         +     indexing_cfg["detector_model"] = detector_model

```

```

328 +         detector = PersonDetector({"indexing": indexing_cfg})
329
330 # Single-pass person detection over ALL candidate windows (open the video ONCE,
read
331 # forward ? no per-window seek). Then compute features per window (pure, no GPU).
332 @@ -95,8 +199,17 @@ def extract_video(
333     state = {"last": t0}
334
335     def _progress(done: int, total: int) -> None:
336 +         rec = None
337         if telemetry is not None:
338             telemetry.snapshot(done, total, phase=spec["name"])
339 +         rec = telemetry.snapshot(done, total, phase=spec["name"])
340 +         if max_gpu_temp_c is not None and rec is not None:
341 +             gpu = rec.get("gpu") or {}
342 +             temp = gpu.get("temp_c")
343 +             if temp is not None and float(temp) >= max_gpu_temp_c:
344 +                 raise RuntimeError(
345 +                     f"{spec['name']}: GPU temperature {float(temp):.1f}C reached "
346 +                     f"--max-gpu-temp-c {max_gpu_temp_c:.1f}; aborting before thermal
shutdown"
347 +                 )
348             now = time.monotonic()
349             if now - state["last"] >= log_every_sec or done >= total:
350                 el = now - t0
351 @@ -160,8 +273,23 @@ def run(
352     iou: float = 0.5,
353     fresh: bool = False,
354     smallest_first: bool = False,
355 +     source_manifest: Optional[str] = None,
356 +     only: Sequence[str] = (),
357 +     missing_fusion_only: bool = False,
358 +     use_gpu: bool = True,
359 +     detector_model: Optional[str] = None,
360 +     max_gpu_temp_c: Optional[float] = None,
361 ) -> List[str]:
362     specs = load_resolved_manifest(manifest_path)
363 +     source_manifest = source_manifest or os.environ.get(SOURCE_MANIFEST_ENV)
364 +     if source_manifest:
365 +         updated = attach_source_video_paths(specs, source_manifest)
366 +         print(
367 +             f"[fusion_golden] attached {updated}/{len(specs)} source video path(s) "
368 +             f"from {source_manifest}",
369 +             flush=True,
370 +         )
371 +     specs = _filter_specs(specs, only)
372     if smallest_first:
373         # Process small videos first for quick feedback (the slow full match lands
last);
374         # the LOVO result is order-independent. Proxy size by the trajectory CSV (one
row/frame).
375 @@ -187,9 +315,19 @@ def run(
376     written: List[str] = []
377     for vi, spec in enumerate(specs, 1):
378         out = os.path.join(out_dir, f"{spec['name']}_golden_features.json")
379 -         # Resumable: skip a video whose JSON already exists (a re-run picks up where a
prior
380 -         # run left off instead of re-detecting). --fresh forces a redo.
381 -         if not fresh and os.path.isfile(out):
382 +         existing = os.path.isfile(out)
383 +         existing_full = existing and is_full_fusion_feature_file(out)
384 +         # Resumable: skip a video whose full-fusion JSON already exists. A leaner
385 +         # windowing-only JSON is stale for A7, so re-extract it even without --fresh.
386 +         if missing_fusion_only and not fresh and existing_full:
387 +             print(

```



```

388 +         f"[fusion_golden] ({vi}/{len(specs)}) {spec['name']}: full-fusion "
389 +         f"features already present -> {out}",
390 +         flush=True,
391 +     )
392 +     written.append(out)
393 +     continue
394 +     if not fresh and existing and existing_full:
395 +         print(
396 +             f"[fusion_golden] ({vi}/{len(specs)}) {spec['name']}: already extracted
"
397 +             f"-> {out} (use --fresh to redo)",
398 @@ -197,6 +335,12 @@ def run(
399 +         )
400 +         written.append(out)
401 +         continue
402 +     if not fresh and existing and not existing_full:
403 +         print(
404 +             f"[fusion_golden] ({vi}/{len(specs)}) {spec['name']}: existing JSON "
405 +             "is not full-fusion schema; re-extracting...",
406 +             flush=True,
407 +         )
408 +         print(
409 +             f"[fusion_golden] ({vi}/{len(specs)}) {spec['name']}: extracting...",
410 +             flush=True,
411 @@ -205,7 +349,15 @@ def run(
412 +         if telem is not None:
413 +             telem.event("video_start", spec["name"])
414 +         try:
415 -             data = extract_video(spec, frame_skip=frame_skip, iou=iou, telemetry=telem)
416 +             data = extract_video(
417 +                 spec,
418 +                 frame_skip=frame_skip,
419 +                 iou=iou,
420 +                 telemetry=telem,
421 +                 use_gpu=use_gpu,
422 +                 detector_model=detector_model,
423 +                 max_gpu_temp_c=max_gpu_temp_c,
424 +             )
425 +         except Exception as e: # noqa: BLE001 - record machine state at death, then
re-raise
426 +             if telem is not None:
427 +                 telem.event("error", spec["name"], err=str(e))
428 @@ -246,6 +398,47 @@ def main() -> None:
429 +         action="store_true",
430 +         help="process smaller videos first for quick feedback (LOVO is order-
independent)",
431 +     )
432 +     ap.add_argument(
433 +         "--source-manifest",
434 +         default=None,
435 +         help=(
436 +             "collector golden_source_videos.json; overrides video paths for
source/proxy "
437 +             f"videos (or set {SOURCE_MANIFEST_ENV})"
438 +         ),
439 +     )
440 +     ap.add_argument(
441 +         "--only",
442 +         action="append",
443 +         default=[],
444 +         metavar="VIDEO_ID",
445 +         help="process one manifest video by name; repeatable",
446 +     )
447 +     ap.add_argument(
448 +         "--missing-fusion-only",

```

```

449 +         action="store_true",
450 +         help="skip videos whose existing JSON already has full shuttle+person fusion
schema",
451 +     )
452 +     ap.add_argument(
453 +         "--cpu",
454 +         action="store_true",
455 +         help="run torchvision person detection on CPU instead of CUDA",
456 +     )
457 +     ap.add_argument(
458 +         "--detector-model",
459 +         default=None,
460 +         choices=[
461 +             "fasterrcnn_resnet50_fpn",
462 +             "fasterrcnn_mobilenet_v3_large_fpn",
463 +             "retinanet_resnet50_fpn",
464 +         ],
465 +         help="torchvision detector model; default keeps the configured Faster R-CNN
ResNet",
466 +     )
467 +     ap.add_argument(
468 +         "--max-gpu-temp-c",
469 +         type=float,
470 +         default=None,
471 +         help="abort extraction if telemetry sees GPU temperature at or above this
value",
472 +     )
473 +     args = ap.parse_args()
474 +     run(
475 +         args.manifest,
476 +         @@ -254,6 +447,12 @@ def main() -> None:
477 +         args.iou,
478 +         fresh=args.fresh,
479 +         smallest_first=args.smallest_first,
480 +         source_manifest=args.source_manifest,
481 +         only=args.only,
482 +         missing_fusion_only=args.missing_fusion_only,
483 +         use_gpu=not args.cpu,
484 +         detector_model=args.detector_model,
485 +         max_gpu_temp_c=args.max_gpu_temp_c,
486 +     )
487
488
489 diff --git a/docs/ALPHA_LAUNCH_READINESS.md b/docs/ALPHA_LAUNCH_READINESS.md
490 index 5f550386..ea59c09b 100644
491 --- a/docs/ALPHA_LAUNCH_READINESS.md
492 +++ b/docs/ALPHA_LAUNCH_READINESS.md
493 @@ -1,6 +1,6 @@
494 # Alpha Launch Readiness Plan
495
496 -> **Status:** `DRAFT` - cross-repo execution tracker; **Owner:** @avidullu;
**Created:** 2026-07-03; **Last updated:** 2026-07-03
497 +> **Status:** `DRAFT` - cross-repo execution tracker; **Owner:** @avidullu;
**Created:** 2026-07-03; **Last updated:** 2026-07-04
498 > **Lifecycle:** `DRAFT` -> IN PROGRESS -> DONE -> archived`
499 > **Tracking anchors:** Section 7 is the progress tracker and source of truth; indexed
in `docs/README.md` and `docs/DOC_STATUS.md`.
500 > **Relation to existing docs:** consolidates the launch-relevant pending items from
`NEXT_STEPS.md`, `ROADMAP.md`, `docs/data_pipeline/GOLDEN_VIDEOS.md`,
`R2_EVAL_METRIC_DESIGN.md`, `COMPUTE_DECOUPLED_SERVING/`, and the sibling `kheIsutra` Studio
docs.
501 @@ -10,7 +10,7 @@
502
503 ## 0. TL;DR
504

```

505 -The corpus is now large enough to move from "grow labels first" into alpha-readiness work, but the 15-video headline is not itself a launch gate pass. `[verified]` The golden manifest has 15 badminton videos; `[analysis]` the local manifest needed path rebasing before manifest-driven commands ran; `[analysis]` only 6 of 15 feature JSONs had the full fusion schema; `[analysis]` the n=15 quality baseline is stale and must be intentionally refreshed before it can gate anything.

506 +The corpus is now large enough to move from "grow labels first" into alpha-readiness work, but the 15-video headline is not itself a launch gate pass. `[verified]` The golden manifest has 15 badminton videos; `[analysis]` A2-A7 made the local manifest/path and full-fusion/audio feature surface reproducible on all 15; `[analysis]` A4/A5 refreshed the n=15 nightly and heuristic floors; `[analysis]` A8/A9 still reject served Gate A default promotion and owned-model serving claims on the current evidence.

507

508 The first controlled alpha should be a hosted, authenticated product loop: upload a video, process it, select rallies, compile, and download a reel. The owned-model/commercial rally-detection claim remains a separate NO-GO until the ship-gate target is ratified and the WASB checkpoint provenance is resolved.

509

510 @@ -31,8 +31,9 @@ This doc separates those into executable workstreams. The goal is to reach an in

511 | D1 | **Product alpha and owned-model launch are separate decisions.** |
`NEXT\_STEPS.md` NO-GO banner, 2026-06-30; verified 2026-07-03 | We can launch a controlled hosted product loop without claiming the owned model is commercially cleared. |

512 | D2 | **First alpha must be page-driven, not SSH/CLI-driven.** | `NEXT\_STEPS.md` P0 UI-driven/shareable item; verified 2026-07-03 | Hosted service, edge auth, upload, job polling, compile, and download are launch-critical. |

513 | D3 | **The 15-video corpus is now a measurement asset, not a free launch pass.** |
`[verified]` `docs/data\_pipeline/GOLDEN\_VIDEOS.md`; `[analysis]` local eval runs, 2026-07-03 | Rebase/portability, feature completeness, baseline refresh, and gate ratification come before quality claims. |

514 -| D4 | **Do not promote served Gate A by default from the current n=15 evidence.** |
`[analysis]` `backend.eval.serve\_contrast` run, 2026-07-03 | Proposal-side lift does not survive served-windowing safety; keep it as an opt-in/analysis lever until owner ratification. |

515 +| D4 | **Do not promote served Gate A by default from the current n=15 evidence.** |
`[analysis]` `backend.eval.serve\_contrast` run, 2026-07-04 | Proposal-side lift does not survive served-windowing safety; keep it as an opt-in/analysis lever until owner ratification. |

516 | D5 | **P10/P11 corpus promotion and train/re-eval are alpha-plus unless owner reopens D13.** | sibling `khehsutra/docs/STUDIO\_MEDIA\_LIFECYCLE.md`; verified 2026-07-03 | First alpha can use manual/offline corpus promotion; automated flywheel follows after serving and gates are stable. |

517 +| D6 | **Gen-0 n=15 shadow training is non-serving.** | `[analysis]`
`training.gen0.harness` run, 2026-07-04 | Learned mean beats the heuristic floor, but the paired sign test is not significant; #172 and WASB C3 remain hard blockers. |

518

519 ## 3. Verified snapshot, 2026-07-03

520

521 @@ -50,10 +51,11 @@ Corpus/eval facts:

522

523 - `[verified]` `docs/data\_pipeline/GOLDEN\_VIDEOS.md` says
`output/human\_lovo\_manifest.json` is the ingested golden corpus at `n = 15`.

524 - `[analysis]` The local manifest had stale absolute paths. A temporary rebased manifest found all 15 trajectories and labels in this workspace.

525 -- `[analysis]` There are 15 `output/\*\_golden\_features.json` files, but only 6 carry full shuttle/person/audio candidate features. The newer 9 are windowing-only schema and are skipped by fusion/person eval.

526 -- `[analysis]` `python -m backend.eval.fusion\_compare --features-dir output` loaded only 6 videos and measured person-fusion `Delta F1 = -0.021`; not promotable.

527 -- `[analysis]` `python -m backend.eval.serve\_contrast --manifest <rebased-n15>` measured proposal Gate A mean `Delta F1 +0.0287`, but served mean `Delta F1 -0.0328`; not served-safe.

528 -- `[analysis]` `python -m training.gen0.harness --manifest <rebased-n15> --floor eval\_baselines/heuristic\_lovo\_n6.json --version analysis-n15 --out <temp-out>` produced learned LOVO mean F1 `0.551`, `ship=False`; this pre-A5 run used the stale n=6 floor, so future comparisons should use `eval\_baselines/heuristic\_lovo\_n15.json`.

529 +- `[analysis]` A7 generated full shuttle/person fusion schema for all 15 videos and A8

audio augmentation populated audio features for all 15.

530 +- `[analysis]` `python -m backend.eval.fusion\_compare --features-dir output` now runs over all 15 and measures person-fusion `Delta F1 = +0.008`; the paired CI crosses zero, so person fusion is not promotable by itself.

531 +- `[analysis]` `python -m backend.eval.fusion\_audio --features-dir output --compare` measures audio `Delta F1 = +0.018`; the paired CI crosses zero, so audio is a shadow/suggestive cue, not a default-flip.

532 +- `[analysis]` `python -m backend.eval.serve\_contrast --manifest output\human\_lovo\_manifest.json` measured proposal Gate A mean `Delta F1 +0.0287`, but served mean `Delta F1 -0.0328`; not served-safe.

533 +- `[analysis]` `python -m training.gen0.harness --manifest output\human\_lovo\_manifest.json --floor eval\_baselines\heuristic\_lovo\_n15.json --version 0.1.1-n15-shadow.20260704 --out artifacts\rally\_tal\_gen0` produced learned LOVO mean F1 `0.551`, floor passed versus heuristic `0.446`, but sign test `9/15`, `p = 0.3036`, `ship=False`.

534 - `[analysis]` `python scripts/nightly\_regression.py --manifest <rebased-n15> --features-dir output --no-report` reported default/milestone tolF1 `0.3636` vs baseline-off tolF1 `0.1382`, but marked the frozen baseline stale due to config/corpus changes.

535 - `[analysis]` Before A6, `python -m backend.eval.ablation --features-dir output --granularity group` failed at import due to a circular import between `backend/eval/ablation.py` and `backend/eval/ablation\_pdf.py`; A6 fixes the module entrypoint and keeps full-fusion ablation scoped to the 6 videos with full schema until A7.

536

537 @@ -97,13 +99,13 @@ Legend: TODO = not started, IN PROGRESS = actively being worked, DONE = shipped/

538 | A0 | Create this alpha readiness tracker and index it in docs | `badminton-highlight-indexer` | - | No | DONE | #451 |

539 | A1 | Stand up alpha serving endpoint: Cloud Run/GPU or approved host, health check, edge auth ON, upload -> process -> jobs -> compile -> download verified | `khelsutra` + `badminton-highlight-indexer` | A0 | Owner spend / CF config | TODO | - |

540 | A2 | Make the n=15 golden manifest portable/reproducible; add a path/corpus smoke check that fails loudly when labels or trajectories are missing | `badminton-highlight-indexer` | A0 | No | DONE | #452 |

541 -| A3 | Promote the 15-video source-video/MD5 records into the collector/vault path; reconcile collector's six-video source manifest with the 15-video eval corpus | `sports-data-collector` + vault | A2 | Owner upload/storage scope | TODO | - |

542 +| A3 | Promote the 15-video source-video/MD5 records into the collector/vault path; reconcile collector's six-video source manifest with the 15-video eval corpus | `sports-data-collector` + vault | A2 | Owner upload/storage scope | IN PROGRESS | collector #62 |

543 | A4 | Refresh the n=15 nightly regression baseline intentionally and record the exact command/output; keep stale-baseline warnings until reviewed | `badminton-highlight-indexer` | A2 | Reviewer sign-off | DONE | #453 |

544 | A5 | Recompute the heuristic n=15 floor and replace the stale `heuristic\_lovo\_n6` floor for future Gen-0 comparisons | `badminton-highlight-indexer` | A2 | No | DONE | #454 |

545 -| A6 | Fix `backend.eval.ablation` CLI circular import, then run the R2/tolF1 ablation on the n=15 corpus | `badminton-highlight-indexer` | A2 | No | IN PROGRESS | this PR |

546 -| A7 | Regenerate or upconvert full-fusion feature JSONs for the newer 9 videos so fusion/person/audio analyses run over all 15 | `badminton-highlight-indexer` | A2 | GPU/data availability | TODO | - |

547 -| A8 | Re-run `fusion\_compare`, group ablation, and served contrast after A7; explicitly promote/reject person fusion, Gate A served-default, and any other R-series default flips | `badminton-highlight-indexer` | A6, A7 | Reviewer sign-off | TODO | - |

548 -| A9 | Run Gen-0/TemporalMaxer shadow training against the refreshed n=15 floor; produce a non-serving artifact bundle and gate verdict | `badminton-highlight-indexer` | A5, A8 | Owner approval for M0.4 scope | TODO | - |

549 +| A6 | Fix `backend.eval.ablation` CLI circular import, then run the R2/tolF1 ablation on the n=15 corpus | `badminton-highlight-indexer` | A2 | No | DONE | #455 |

550 +| A7 | Regenerate or upconvert full-fusion feature JSONs for the newer 9 videos so fusion/person/audio analyses run over all 15 | `badminton-highlight-indexer` | A2 | GPU/data availability | IN PROGRESS | this PR |

551 +| A8 | Re-run `fusion\_compare`, group ablation, and served contrast after A7; explicitly promote/reject person fusion, Gate A served-default, and any other R-series default flips | `badminton-highlight-indexer` | A6, A7 | Reviewer sign-off | IN PROGRESS | this PR |

552 +| A9 | Run Gen-0/TemporalMaxer shadow training against the refreshed n=15 floor; produce a non-serving artifact bundle and gate verdict | `badminton-highlight-indexer` | A5, A8 | Owner approval for M0.4 scope | IN PROGRESS | this PR |

```

553 | A10 | Decide and record the alpha ship-gate target: metric, corpus slice, minimum
threshold, significance rule, and what claims are allowed | Owner + `badminton-highlight-
indexer` | A4, A5, A8 | Owner decision | GATED | - |
554 | A11 | Resolve or explicitly defer WASB C3 provenance for alpha messaging; if
deferred, document "no commercial model sharing" copy guardrail | Owner/legal | A10 |
Legal/provenance | GATED | - |
555 | A12 | Reopen or defer P10/P11 corpus flywheel: `PATCH /api/segments`, collector
promotion, `train`/`reeval` job, delta-F1 surface, and alignment with
`COMPUTE_DECOUPLED_SERVING` M11 | `khesutra` + `badminton-highlight-indexer` + collector | A1,
A3 | Owner D13 reopen (`khesutra/docs/STUDIO_MEDIA_LIFECYCLE.md`) | GATED | - |
556 @@ -174,22 +176,57 @@ python -m backend.eval.rally_seg_eval --manifest
output\human_lovo_manifest.json
557
558 Result: the ablation CLI now runs, but full-fusion feature ablation still reports `n=6`
because only the original six feature JSONs carry the full fusion schema; A7 must
regenerate/upconvert the newer nine before fusion/person/audio promotion decisions. The n=15
R2/tolF1 contrast is positive for the served milestone versus baseline-off: baseline/off `tolF1
= 0.138`, served `tolF1 = 0.364`, `Delta tolF1 = +0.225` with CI `[+0.165, +0.283]`, sign
`14/0`, `p = 0.000`; `Delta F1@0.5 = +0.236` with CI `[+0.161, +0.313]`; merge rate improves
from `0.638` to `0.135`. Net over-segmentation guard passes (`1.276 -> 0.378`), while the strict
segment-count rule remains blocked by increased splits, so R2/tolF1 is the supported gate for
the baseline-off to served milestone contrast rather than a blanket fusion/default-flip
approval.
559
560 +A7 full-fusion/audio feature refresh record:
561 +
562 +```powershell
563 +python -m backend.eval.fusion_golden --manifest output\human_lovo_manifest.json --out-
dir output --source-manifest ..\sports-data-collector\data\golden_source_videos.json --missing-
fusion-only --smallest-first --max-gpu-temp-c 87
564 +python -m backend.eval.fusion_audio --manifest output\human_lovo_manifest.json
--features-dir output --source-manifest ..\sports-data-collector\data\golden_source_videos.json
--augment
565 +```
566 +
567 +Result: the GPU pass completed without a thermal abort after fans were set to max
performance. Full-fusion schema is now present for `15/15` feature JSONs, and audio features are
present for `15/15`. The large regenerated clips completed as follows: `mahadevpura_1` `1`
window / `0` positives / `10` GT, `GX020094` `99` / `30` / `40`, `GX030094` `79` / `21` / `41`,
`GX010141` `26` / `11` / `29`, `GX010137` `56` / `31` / `34`, `Boxhill_Doubles` `59` / `12` /
`43`, `mahadevpura_2` `14` / `2` / `40`, `GX010128` `51` / `15` / `52`, `Badminton_BXH_2` `29` /
`1` / `44`. The source manifest supplied all 15 source-video paths, including the F-drive
Boxhill originals.
568 +
569 +A8 fusion/default-decision record:
570 +
571 +```powershell
572 +python -m backend.eval.fusion_compare --features-dir output
573 +python -m backend.eval.fusion_audio --features-dir output --compare
574 +python -m backend.eval.ablation --features-dir output --granularity group
575 +python -m backend.eval.serve_contrast --manifest output\human_lovo_manifest.json
576 +python -m backend.eval.rally_seg_eval --manifest output\human_lovo_manifest.json
--features-dir output --preset served --max-window-duration 0 --inactivity-end 0 --vs served
--vs-max-window-duration 6 --vs-inactivity-end 1.5 --json-out output\A6_r2_ablation.json
577 +```
578 +
579 +Result:
580 +
581 +- Person fusion is **not promoted**: shuttle-only F1 `0.302`, shuttle+person F1
`0.311`, `Delta F1 +0.008`, paired CI `[-0.040, +0.055]`, sign `8W/6L`, `p = 0.791`.
582 +- Audio is **shadow/suggestive, not promoted**: shuttle+person F1 `0.311`,
shuttle+person+audio F1 `0.329`, `Delta F1 +0.018`, paired CI `[-0.017, +0.053]`, sign `10W/4L`,
`p = 0.180`.
583 +- Group ablation remains non-significant for cue promotion: audio `+0.018` CI `[-0.017,
+0.053]`; shuttle `+0.013` CI `[-0.002, +0.029]`; gate_a `-0.001` CI `[-0.012, +0.008]`; person

```

```

~-0.009` CI `[-0.045, +0.021]`.
584   +- Gate A served default is **rejected**: proposal mean `Delta F1 +0.0287`, CI
`[-0.0222, +0.0680]`; served mean `Delta F1 -0.0328`, CI `[-0.0799, +0.0052]`, `regresses=True`.
585   +- R2/cap/inactivity remains the strong quality result: served candidate F1 `0.472` vs
baseline-off `0.237`, `Delta F1 +0.236`, CI `[+0.161, +0.313]`; tolF1 `0.364` vs `0.138`, `Delta
tolF1 +0.225`, CI `[+0.165, +0.283]`; merge rate `0.135` vs `0.638`. Net over-seg guard passes,
strict per-video split-count guard still blocks blanket promotion language.
586   +
587   +A9 Gen-0 shadow-training record:
588   +
589   +```powershell
590   +python -m training.gen0.harness --manifest output\human_lovo_manifest.json --floor
eval_baselines\heuristic_lovo_n15.json --out artifacts\rally_tal_gen0 --version
0.1.1-n15-shadow.20260704
591   +```
592   +
593   +Result: current Gen-0 LogReg/TAL harness produced local bundle
`artifacts\rally_tal_gen0\0.1.1-n15-shadow.20260704` with learned LOVO mean F1 `0.551` versus
heuristic floor `0.4456`, so `floor_passed=True`. The gate remains **non-serving**: paired sign
test `9/15`, `p = 0.3036`, `ship=False`; blockers are uncalibrated ship-gate target #172 and
WASB checkpoint provenance C3. TemporalMaxer itself is not implemented in this branch; the
evidence supports either closing A9 as a Gen-0 shadow baseline or opening a follow-up M0.4
scorer-swap PR.
594   +
595   ## 9. Open questions
596
597   1. **Owner:** Is the first alpha allowed to use manual/offline corpus promotion, or
should P10/P11 be reopened before any tester wave?
598   2. **Owner/platform:** Which hosted compute target is the next stand-up: Cloud Run GPU,
GCE L4 VM, or another approved GPU host?
599   3. **Owner/legal:** What is the acceptable alpha wording while WASB C3 remains
unresolved?
600   4. **Engineering:** Should the portable corpus manifest live as a generated artifact, a
checked-in redacted manifest, or both?
601   -5. **Engineering:** Should n=15 feature regeneration target full fusion schema for
every clip, or should the eval loaders support mixed schemas with explicit capability tags?
602   +5. **Engineering:** Should A9 close on the current Gen-0 non-serving bundle, or should
the actual TemporalMaxer scorer swap be implemented before marking A9 done?
603
604   ## 10. Definition of done
605
606   - [ ] Invited tester can complete the hosted record/upload -> reel download CUJ without
SSH.
607   - [ ] Alpha service is gated behind the chosen auth posture and has a documented
disable/rollback path.
608   -- [ ] n=15 corpus is portable enough that manifest-driven quality commands do not rely
on stale local absolute paths.
609   -- [ ] n=15 nightly baseline and heuristic floor are refreshed with recorded commands.
610   -- [ ] R2/tolF1 gate is promoted or rejected with a recorded ablation result.
611   -- [ ] Fusion/person/Gate-A served-default decisions are explicitly promoted or rejected
on the 15-video corpus.
612   +- [x] n=15 corpus is portable enough that manifest-driven quality commands do not rely
on stale local absolute paths.
613   +- [x] n=15 nightly baseline and heuristic floor are refreshed with recorded commands.
614   +- [x] R2/tolF1 gate is promoted or rejected with a recorded ablation result.
615   +- [x] Fusion/person/Gate-A served-default decisions are explicitly promoted or rejected
on the 15-video corpus.
616   - [ ] Ship-gate target and allowed alpha claims are recorded.
617   - [ ] WASB C3 status is recorded and reflected in alpha messaging.
618   - [ ] P10/P11 are either reopened with a committed sequence or explicitly deferred for
alpha-plus.
619   @@ -226,3 +263,4 @@ python -m backend.eval.rally_seg_eval --manifest
output\human_lovo_manifest.json
620   ### Changelog
621

```

```

622     - 2026-07-03 - Created from a synced five-repo review of `NEXT_STEPS`, `ROADMAP`,
golden-corpus docs, local eval artifacts, and the current 15-video corpus state.
623     +- 2026-07-04 - Recorded A7 full-fusion/audio regeneration, A8 n15 fusion/default
decisions, and A9 Gen-0 n15 shadow-training gate verdict.
624     diff --git a/tests/test_ablation.py b/tests/test_ablation.py
625     index a4491d7c..b5450f7b 100644
626     --- a/tests/test_ablation.py
627     +++ b/tests/test_ablation.py
628     @@ -10,9 +10,10 @@
629     - the honest-stats contract ? every row ships a CI + sign test, and no verdict is
stronger
630         than the stats support (the no-over-claim contract).
631     - a valid non-trivial PDF is produced (asserted via pypdf page count > 0).
632     - - the golden litmus reproduction ? on the 6 real ``output/*_golden_features.json``
the runner
633     - reproduces the hand-measured Gate-A result (?F1 ? +0.074, 6/6 wins, p ? 0.031).
Skips
634     - cleanly when the golden JSONs are absent (CI), runs when present (owner box).
635     + - the golden litmus reproduction ? on real ``output/*_golden_features.json`` files
the runner
636     + reproduces the hand-measured Gate-A result for either the legacy n=6 substrate or
the refreshed
637     + n=15 A7 substrate. Skips cleanly when the golden JSONs are absent (CI), runs when
present
638     + (owner box).
639     """
640
641     from __future__ import annotations
642     @@ -517,7 +518,7 @@ def test_load_videos_skips_windowing_only_schema(tmp_path):
643
644
645     def _golden_present() -> bool:
646     - """True iff ?6 FULL-FUSION-SCHEMA golden files exist (the litmus needs the
shuttle/person cue
647     + """True iff >=6 FULL-FUSION-SCHEMA golden files exist (the litmus needs the
shuttle/person cue
648         columns). Counts only ablatable files ? leaner velocity-windowing files are skipped
by
649         ``load_videos``, so counting them would trip the litmus on an un-ablatable corpus
(the n=11
650         mixed-schema bug)."""
651     @@ -540,29 +541,42 @@ def _golden_present() -> bool:
652         reason="golden output/*_golden_features.json absent (run fusion_golden first)",
653     )
654     def test_litmus_reproduces_gate_a():
655     - """LITMUS: on the 6 real golden feature JSONs the runner reproduces the hand-
measured Gate-A
656     - result ? marginal ?F1 ? +0.074, 6/6 wins, p ? 0.031, CI ? [+0.042, +0.104], within
tolerance.
657     - This validates the whole framework against the recorded ground truth. Skips cleanly
in CI
658     - where the golden substrate is absent; runs on the owner box where it is present."""
659     + """LITMUS: on real golden feature JSONs the runner reproduces the recorded Gate-A
result.
660     +
661     + The legacy n=6 owner-box substrate preserves the 2026-06-14 hand measurement. After
A7, the
662     + refreshed owner-box substrate has n=15 full-fusion files; that result is
intentionally weaker
663     + and supports "keep structural, do not promote served default".
664     + """
665         videos = ablation.load_videos(GOLDEN_DIR)
666     - assert len(videos) == 6
667     + assert len(videos) in {6, 15}
668         cfg = ablation.litmus_config()

```

```

669         report = run_ablation(videos, cfg)
670         assert len(report.rows) == 1
671         row = report.rows[0]
672         assert row.signal == "gate_a"
673         - # The recorded QUALITY_ITERATIONS truth (2026-06-14), reproduced within tolerance.
674         - assert row.delta_f1 == pytest.approx(0.074, abs=0.005)
675         - assert row.sign_wins == 6 and row.sign_losses == 0
676         - assert row.sign_p_value == pytest.approx(0.031, abs=0.005)
677         - assert row.ci_low == pytest.approx(0.042, abs=0.01)
678         - assert row.ci_high == pytest.approx(0.104, abs=0.01)
679         - assert row.ci_excludes_zero is True
680         + if len(videos) == 6:
681         +     # The recorded QUALITY_ITERATIONS truth (2026-06-14), reproduced within
tolerance.
682         +     assert row.delta_f1 == pytest.approx(0.074, abs=0.005)
683         +     assert row.sign_wins == 6 and row.sign_losses == 0
684         +     assert row.sign_p_value == pytest.approx(0.031, abs=0.005)
685         +     assert row.ci_low == pytest.approx(0.042, abs=0.01)
686         +     assert row.ci_high == pytest.approx(0.104, abs=0.01)
687         +     assert row.ci_excludes_zero is True
688         +     # C4: over-segmentation drops 1.68 -> 1.01 when Gate-A is on (full).
689         +     assert row.seg_ratio_full == pytest.approx(1.01, abs=0.05)
690         +     assert row.seg_ratio_ablated == pytest.approx(1.68, abs=0.05)
691         + else:
692         +     # The A7 refreshed n=15 substrate: proposal-side structural signal, but CI
crosses zero.
693         +     assert row.delta_f1 == pytest.approx(0.0287, abs=0.005)
694         +     assert row.sign_wins == 12 and row.sign_losses == 1
695         +     assert row.sign_p_value == pytest.approx(0.0034, abs=0.002)
696         +     assert row.ci_low == pytest.approx(-0.020, abs=0.01)
697         +     assert row.ci_high == pytest.approx(0.066, abs=0.01)
698         +     assert row.ci_excludes_zero is False
699         +     assert row.seg_ratio_full == pytest.approx(0.79, abs=0.05)
700         +     assert row.seg_ratio_ablated == pytest.approx(1.36, abs=0.05)
701         +     # Gate-A is structural ? reported but never auto-promoted to earns_keep.
702         +     assert row.verdict == ablation.STRUCTURAL_PROTECTED
703         - # C4: over-segmentation drops 1.68 ? 1.01 when Gate-A is on (full).
704         - assert row.seg_ratio_full == pytest.approx(1.01, abs=0.05)
705         - assert row.seg_ratio_ablated == pytest.approx(1.68, abs=0.05)
706
707
708     @pytest.mark.skipif(
709     @@ -572,7 +586,7 @@ def test_litmus_report_is_well_formed():
710         """The litmus report is a proper AblationReport (provenance + a sorted single
row)."""
711         report = run_ablation(ablation.load_videos(GOLDEN_DIR), ablation.litmus_config())
712         assert isinstance(report, AblationReport)
713         - assert report.num_videos == 6
714         + assert report.num_videos in {6, 15}
715         assert report.full_set_signals == ["gate_a"]
716         assert report.label_set_hash
717         out = ablation.format_table(report)
718         diff --git a/tests/test_audio_features.py b/tests/test_audio_features.py
719         index 26028591..d962b5e4 100644
720         --- a/tests/test_audio_features.py
721         +++ b/tests/test_audio_features.py
722         @@ -118,6 +118,7 @@ def
test_augment_rebases_stale_manifest_before_deriving_video_path(tmp_path, mon
723             out.mkdir(parents=True)
724             traj.write_text("x\n", encoding="utf-8")
725             labels.write_text("start_time,end_time\n0,1\n", encoding="utf-8")
726             + labels.with_name("Foo.MP4").write_text("video\n", encoding="utf-8")
727             manifest = out / "human_lovo_manifest.json"
728             manifest.write_text(
729                 json.dumps(

```



```

730 @@ -163,6 +164,80 @@ def _fake_detect(video_path, k=2.5):
731     assert data["candidates"][0]["audio"]["audio_hit_count"] == 1.0
732
733
734 +def test_augment_uses_source_manifest_video_path(tmp_path, monkeypatch):
735 +    workspace = tmp_path / "workspace"
736 +    repo = workspace / "badminton-highlight-indexer"
737 +    out = repo / "output"
738 +    traj = workspace / "Annotation Setup" / "Trajectories" / "foo.csv"
739 +    labels = workspace / "Annotation Setup" / "Golden Labelled" / "Foo.rallies.csv"
740 +    source_video = workspace / "source" / "Foo_proxy.mp4"
741 +    traj.parent.mkdir(parents=True)
742 +    labels.parent.mkdir(parents=True)
743 +    source_video.parent.mkdir(parents=True)
744 +    out.mkdir(parents=True)
745 +    traj.write_text("x\n", encoding="utf-8")
746 +    labels.write_text("start_time,end_time\n0,1\n", encoding="utf-8")
747 +    source_video.write_text("video\n", encoding="utf-8")
748 +    manifest = out / "human_lovo_manifest.json"
749 +    manifest.write_text(
750 +        json.dumps(
751 +            [
752 +                {
753 +                    "name": "foo",
754 +                    "trajectory": str(traj),
755 +                    "labels": str(labels),
756 +                    "fps": 30.0,
757 +                    "frame_width": 1920.0,
758 +                }
759 +            ],
760 +            encoding="utf-8",
761 +        )
762 +    )
763 +    source_manifest = workspace / "golden_source_videos.json"
764 +    source_manifest.write_text(
765 +        json.dumps(
766 +            {
767 +                "schema": "sports-data-collector.golden_source_videos.v2",
768 +                "videos": [
769 +                    {
770 +                        "video_id": "foo",
771 +                        "local_path": str(source_video),
772 +                        "source_status": "present_local",
773 +                    }
774 +                ],
775 +            },
776 +            encoding="utf-8",
777 +        )
778 +    )
779 +    feature_json = out / "foo_golden_features.json"
780 +    feature_json.write_text(
781 +        json.dumps(
782 +            {
783 +                "name": "foo",
784 +                "fps": 30.0,
785 +                "frame_width": 1920.0,
786 +                "candidates": [{"start": 0.0, "end": 1.0}],
787 +                "gts": [[0.0, 1.0]],
788 +            },
789 +            encoding="utf-8",
790 +        )
791 +    )
792 +    seen = []
793 +
794 +    def _fake_detect(video_path, k=2.5):

```

```

795     +         seen.append((video_path, k))
796     +         return [H(0.5, 3.0)]
797     +
798     +     monkeypatch.setattr(fusion_audio, "detect_hits_in_video", _fake_detect)
799     +
800     +     written = fusion_audio.augment(
801     +         str(manifest), str(out), k=1.5, source_manifest=str(source_manifest)
802     +     )
803     +
804     +     assert written == [str(feature_json)]
805     +     assert seen == [(str(source_video), 1.5)]
806     +
807     +
808     def test_audio_increment_compare(tmp_path):
809         d = str(tmp_path)
810         for k in range(6):
811             diff --git a/tests/test_fusion_compare.py b/tests/test_fusion_compare.py
812             index f042eea1..6b4226e6 100644
813             --- a/tests/test_fusion_compare.py
814             +++ b/tests/test_fusion_compare.py
815             @@ -181,6 +181,7 @@ def detections_over_windows(self, video, windows, frame_skip,
progress_cb=None):
816                 "fps": 30.0,
817                 "frame_width": 1920.0,
818                 "labels": "foo.rallies.csv",
819             +         "video_path": __file__,
820             }
821             data = fusion_golden.extract_video(
822                 spec, detector=_StubDetector(), frame_skip=3, iou=0.5
823             @@ -197,6 +198,47 @@ def detections_over_windows(self, video, windows, frame_skip,
progress_cb=None):
824                 assert all(isinstance(v, float) for v in c0["person"].values())
825
826
827     +def test_extract_video_aborts_on_max_gpu_temp(monkeypatch):
828     +     from backend.eval import fusion_golden
829     +     from backend.pipeline.detectors.tracknet_runner import TrackNetRunner
830     +
831     +     monkeypatch.setattr(
832     +         fusion_golden.distill_local,
833     +         "build_candidates",
834     +         lambda traj, fps, fw: ([[(0.0, 1.0)], [[1.0, 0.1, 0.1, 0.1, 0.0]]]),
835     +     )
836     +     monkeypatch.setattr(
837     +         TrackNetRunner, "parse_trajectory_csv", staticmethod(lambda traj: [])
838     +     )
839     +     monkeypatch.setattr(fusion_golden, "load_golden_gts", lambda labels: [])
840     +
841     +     class _HotTelemetry:
842     +         def snapshot(self, done, total, phase=None):
843     +             return {"gpu": {"temp_c": 90.0}}
844     +
845     +     class _StubDetector:
846     +         def detections_over_windows(self, video, windows, frame_skip,
progress_cb=None):
847             +             progress_cb(1, 10)
848             +             return []
849             +
850             +     spec = {
851             +         "name": "hot",
852             +         "trajectory": "bar.csv",
853             +         "fps": 30.0,
854             +         "frame_width": 1920.0,
855             +         "labels": "foo.rallies.csv",
856             +         "video_path": __file__,

```

```

857 +     }
858 +
859 +     with pytest.raises(RuntimeError, match="GPU temperature 90.0C"):
860 +         fusion_golden.extract_video(
861 +             spec,
862 +             detector=_StubDetector(),
863 +             telemetry=_HotTelemetry(),
864 +             max_gpu_temp_c=85.0,
865 +         )
866 +
867 +
868     def test_fusion_golden_run_is_resumable(tmp_path, monkeypatch):
869         """run() skips a video whose JSON already exists (resume a long run); --fresh
870         redoes all."""
871         from backend.eval import fusion_golden
872         @@ -261,6 +303,121 @@ def _stub(spec, **kw):
873         assert extracted == ["alpha", "beta"] # --fresh redoes both
874
875     +def test_fusion_golden_run_reextracts_windowing_only_json(tmp_path, monkeypatch):
876     +    """A7: an existing lean windowing JSON is stale for fusion and must be
877     +    regenerated."""
878     +    from backend.eval import fusion_golden
879     +
880     +    out_dir = str(tmp_path / "out")
881     +    os.makedirs(out_dir)
882     +    (tmp_path / "a.csv").write_text("x\n", encoding="utf-8")
883     +    (tmp_path / "a.rallies.csv").write_text("x\n", encoding="utf-8")
884     +    manifest = [
885     +        {
886     +            "name": "alpha",
887     +            "trajectory": "a.csv",
888     +            "fps": 30.0,
889     +            "frame_width": 1920.0,
890     +            "labels": "a.rallies.csv",
891     +        }
892     +    ]
893     +    manifest_path = str(tmp_path / "m.json")
894     +    with open(manifest_path, "w", encoding="utf-8") as f:
895     +        json.dump(manifest, f)
896     +    _write_windowing_video(out_dir, "alpha")
897     +
898     +    extracted = []
899     +
900     +    def _stub(spec, **kw):
901     +        extracted.append(spec["name"])
902     +        return {
903     +            "name": spec["name"],
904     +            "fps": 30.0,
905     +            "frame_width": 1920.0,
906     +            "candidates": [],
907     +            "gts": [],
908     +        }
909     +
910     +    monkeypatch.setattr(fusion_golden, "extract_video", _stub)
911     +
912     +    fusion_golden.run(manifest_path, out_dir, fresh=False)
913     +
914     +    assert extracted == ["alpha"]
915     +    assert fusion_golden.is_full_fusion_feature_file(
916     +        os.path.join(out_dir, "alpha_golden_features.json")
917     +    )
918     +
919     +def test_fusion_golden_source_manifest_supplies_video_paths(tmp_path, monkeypatch):

```

```

920 +     """The collector source manifest can provide Drive/F-drive video paths for A7
921 +     extraction."""
922 +
923 +     from backend.eval import fusion_golden
924 +
925 +     out_dir = str(tmp_path / "out")
926 +     os.makedirs(out_dir)
927 +     for name in ("a.csv", "b.csv", "a.rallies.csv", "b.rallies.csv"):
928 +         (tmp_path / name).write_text("\n", encoding="utf-8")
929 +     source_video = tmp_path / "source-alpha.mp4"
930 +     source_video.write_text("video\n", encoding="utf-8")
931 +     source_manifest = tmp_path / "golden_source_videos.json"
932 +     source_manifest.write_text(
933 +         json.dumps(
934 +             {
935 +                 "schema": "sports-data-collector.golden_source_videos.v2",
936 +                 "videos": [
937 +                     {
938 +                         "video_id": "alpha",
939 +                         "local_path": str(source_video),
940 +                         "source_status": "present_local",
941 +                     },
942 +                     {
943 +                         "video_id": "beta",
944 +                         "local_path": str(tmp_path / "missing.mp4"),
945 +                         "source_status": "missing_local_source_hash_from_run_log",
946 +                     },
947 +                 ],
948 +             },
949 +             encoding="utf-8",
950 +         )
951 +     )
952 +     manifest = [
953 +         {
954 +             "name": "alpha",
955 +             "trajectory": "a.csv",
956 +             "fps": 30.0,
957 +             "frame_width": 1920.0,
958 +             "labels": "a.rallies.csv",
959 +             "video_path": str(tmp_path / "stale-alpha.mp4"),
960 +         },
961 +         {
962 +             "name": "beta",
963 +             "trajectory": "b.csv",
964 +             "fps": 30.0,
965 +             "frame_width": 1920.0,
966 +             "labels": "b.rallies.csv",
967 +         },
968 +     ]
969 +     manifest_path = str(tmp_path / "m.json")
970 +     with open(manifest_path, "w", encoding="utf-8") as f:
971 +         json.dump(manifest, f)
972 +
973 +     seen = []
974 +
975 +     def _stub(spec, **kw):
976 +         seen.append((spec["name"], spec.get("video_path")))
977 +         return {
978 +             "name": spec["name"],
979 +             "fps": 30.0,
980 +             "frame_width": 1920.0,
981 +             "candidates": [],
982 +             "gts": [],
983 +         }
984 +
985 +     monkeypatch.setattr(fusion_golden, "extract_video", _stub)

```

```

984 +
985 + fusion_golden.run(manifest_path, out_dir, source_manifest=str(source_manifest))
986 +
987 + assert seen == [("alpha", str(source_video)), ("beta", None)]
988 +
989 +
990 def test_fusion_golden_smallest_first(tmp_path, monkeypatch):
991     """--smallest-first orders by trajectory size; default keeps manifest order."""
992     from backend.eval import fusion_golden
993

```

6. assistant (2026-07-04T00:45:13.999Z)

Now let me read the scratchpad files to examine the code more closely:

7. user (2026-07-04T00:45:14.646Z)

```

1     """Offline golden-set person-feature extractor ? the fusion-P3 measurement's GPU step.
2
3     Mirrors `distill_local`'s CSV-driven flow but ADDS the ~5 person-cue features per
shuttle
4     candidate window, by running `PersonDetector` over the ORIGINAL video. It reuses the
5     EXISTING golden trajectory CSVs (so **no WASB re-run**), needs **no Gemini and no DB**,
and
6     writes one small JSON per video. The LOVO comparison (`fusion_compare.py`) then runs
100% on
7     CPU offline. So the GPU touches exactly one thing ? person detection ? and the lift is
8     measured by the shipped experiment harness; person features enter only as feature
columns the
9     learned scorer weights (no special-casing, default-OFF until the golden ?F1 justifies
them).
10
11     Run (on the GPU box):
12         python -m backend.eval.fusion_golden --manifest output/human_lovo_manifest.json
--out-dir output
13
14     Each output `output/<name>_golden_features.json` is the replayable feature substrate ?
re-run
15     `fusion_compare` with different variants/thresholds forever without re-detecting on the
GPU.
16     """
17
18     from __future__ import annotations
19
20     import argparse
21     import json
22     import os
23     import time
24     from typing import Any, Dict, List, Optional, Sequence
25
26     from backend.eval import distill_local
27     from backend.eval.calibrate_wasb import load_golden_gts
28     from backend.eval.golden_manifest import load_resolved_manifest
29     from backend.eval.training import label_candidates
30     from backend.pipeline.detectors import fusion_features as ff
31     from backend.pipeline.detectors import rally_gate
32     from backend.pipeline.detectors.tracknet_runner import TrackNetRunner
33     from backend.utils.run_telemetry import RunTelemetry
34
35     # The 5 fps/resolution-invariant shuttle features distill_local already computes.
36     SHUTTLE_FEATURE_NAMES = list(distill_local.FEATURE_NAMES)
37     # The 5 person-cue features (fusion_features.PERSON_FEATURE_NAMES).
38     PERSON_FEATURE_NAMES = list(ff.PERSON_FEATURE_NAMES)
39     SOURCE_MANIFEST_ENV = "RALLY_GOLDEN_SOURCE_MANIFEST"
40

```

```

41
42 def is_full_fusion_feature_file(path: str) -> bool:
43     """True when an existing golden feature JSON already has shuttle+person cue dicts.
44
45     Corpus-growth runs can leave leaner windowing-only JSONs with the same filename
46     pattern.
47     Those are useful for gts/windowing analysis, but they are not the replayable fusion
48     substrate that ``fusion_compare`` and ``ablation`` need.
49     """
50     try:
51         with open(path, encoding="utf-8") as f:
52             data = json.load(f)
53     except (OSError, json.JSONDecodeError, TypeError):
54         return False
55     candidates = data.get("candidates")
56     if not isinstance(candidates, list):
57         return False
58     if not candidates:
59         return True
60     return all(
61         isinstance(candidate, dict)
62         and "shuttle" in candidate
63         and "person" in candidate
64         for candidate in candidates
65     )
66
67
68 def load_source_video_paths(source_manifest_path: Optional[str]) -> Dict[str, str]:
69     """Load ``video_id -> local_path`` from the collector golden-source manifest.
70
71     The collector records source/proxy ownership and full-file MD5s. Only entries marked
72     ``present_local`` with a concrete ``local_path`` can drive local GPU extraction.
73     """
74
75     if not source_manifest_path:
76         return {}
77     with open(source_manifest_path, encoding="utf-8") as f:
78         data = json.load(f)
79     entries = data.get("videos", data) if isinstance(data, dict) else data
80     if not isinstance(entries, list):
81         raise ValueError(f"source manifest must contain a videos list: {source_manifest_path}")
82
83     paths: Dict[str, str] = {}
84     for entry in entries:
85         if not isinstance(entry, dict):
86             continue
87         video_id = entry.get("video_id") or entry.get("name")
88         local_path = entry.get("local_path")
89         if not video_id or not local_path:
90             continue
91         if entry.get("source_status", "present_local") != "present_local":
92             continue
93         paths[str(video_id)] = str(local_path)
94     return paths
95
96
97 def attach_source_video_paths(
98     specs: Sequence[Dict[str, Any]], source_manifest_path: Optional[str]
99 ) -> int:
100     """Attach explicit ``video_path`` fields to manifest specs from the source manifest.
101
102     Returns the number of specs updated. The collector source manifest is authoritative
103     for local source/proxy video location and overrides derived/stale manifest paths.

```

```

104     """
105
106     source_paths = load_source_video_paths(source_manifest_path)
107     updated = 0
108     for spec in specs:
109         name = str(spec.get("name", ""))
110         source_path = source_paths.get(name)
111         if not source_path:
112             continue
113         if spec.get("video_path") == source_path:
114             continue
115         spec["video_path"] = source_path
116         updated += 1
117     return updated
118
119
120 def _filter_specs(
121     specs: Sequence[Dict[str, Any]], only: Sequence[str]
122 ) -> List[Dict[str, Any]]:
123     if not only:
124         return list(specs)
125     wanted = list(dict.fromkeys(only))
126     wanted_set = set(wanted)
127     filtered = [spec for spec in specs if str(spec.get("name")) in wanted_set]
128     found = {str(spec.get("name")) for spec in filtered}
129     missing = [name for name in wanted if name not in found]
130     if missing:
131         raise ValueError(f"--only requested unknown video(s): {' '.join(missing)}")
132     return filtered
133
134
135 def _derive_video_path(spec: Dict[str, Any]) -> str:
136     """Original MP4 for a manifest entry ? explicit `video_path`, else derived from the
137     golden labels path (`X.rallies.csv` -> `X.MP4`, the golden-set naming
138     convention)."""
139     if spec.get("video_path"):
140         return str(spec["video_path"])
141     labels = str(spec["labels"])
142     if labels.endswith(".rallies.csv"):
143         return labels[: -len(".rallies.csv")] + ".MP4"
144     raise ValueError(
145         f"{spec.get('name')}: no video_path and labels {labels!r} isn't *.rallies.csv"
146     )
147
148 def extract_video(
149     spec: Dict[str, Any],
150     detector: Optional[Any] = None,
151     frame_skip: int = 3,
152     iou: float = 0.5,
153     log_every_sec: float = 15.0,
154     telemetry: Optional[Any] = None,
155     use_gpu: bool = True,
156     detector_model: Optional[str] = None,
157     max_gpu_temp_c: Optional[float] = None,
158 ) -> Dict[str, Any]:
159     """One golden video -> a feature record (shuttle + person features per candidate
160     window).
161
162     Returns `{name, fps, frame_width,
163     candidates:[{start,end,shuttle{ },person{ },label}], gts}`.
164     `detector` (anything with `detections_over_windows`) is injectable for GPU-free
165     tests;
166
167     when None a real `PersonDetector` is built (loads torchvision on first use).
168     `telemetry` (a `RunTelemetry` or None) records per-frame velocity + machine-health

```

```

snapshots
165     from the detection progress callback so a hard kill leaves a forensic runlog.
166     """
167     fps, fw = float(spec["fps"]), float(spec["frame_width"])
168     traj = str(spec["trajectory"])
169     points = TrackNetRunner.parse_trajectory_csv(traj)
170     intervals, x_shuttle = distill_local.build_candidates(traj, fps, fw)
171     gts = load_golden_gts(str(spec["labels"]))
172     labels = label_candidates(intervals, gts, iou)
173     # Net axis for the person two-sided split ? reuse rally_gate's camera-agnostic
estimate
174     # (parity with the shuttle net-crossing gate) rather than hard-coding an
orientation.
175     axis, net_px, _span = rally_gate.estimate_play_axis(points)
176     video_path = _derive_video_path(spec)
177     if not os.path.isfile(video_path):
178         raise FileNotFoundError(
179             f"{spec.get('name')}: source video not found for fusion extraction:
{video_path}"
180         )
181
182     if detector is None:
183         from backend.pipeline.detectors.person_detector import PersonDetector
184
185         indexing_cfg: Dict[str, Any] = {"use_gpu": use_gpu}
186         if detector_model:
187             indexing_cfg["detector_model"] = detector_model
188         detector = PersonDetector({"indexing": indexing_cfg})
189
190     # Single-pass person detection over ALL candidate windows (open the video ONCE, read
191     # forward ? no per-window seek). Then compute features per window (pure, no GPU).
192     n = len(intervals)
193     win_frames = [(round(s * fps), round(e * fps)) for (s, e) in intervals]
194     print(
195         f"[fusion_golden]   {spec['name']}: {n} windows ? person-detecting in one
pass...",
196         flush=True,
197     )
198     t0 = time.monotonic()
199     state = {"last": t0}
200
201     def _progress(done: int, total: int) -> None:
202         rec = None
203         if telemetry is not None:
204             rec = telemetry.snapshot(done, total, phase=spec["name"])
205         if max_gpu_temp_c is not None and rec is not None:
206             gpu = rec.get("gpu") or {}
207             temp = gpu.get("temp_c")
208             if temp is not None and float(temp) >= max_gpu_temp_c:
209                 raise RuntimeError(
210                     f"{spec['name']}: GPU temperature {float(temp):.1f}C reached "
211                     f"--max-gpu-temp-c {max_gpu_temp_c:.1f}; aborting before thermal
shutdown"
212                 )
213         now = time.monotonic()
214         if now - state["last"] >= log_every_sec or done >= total:
215             el = now - t0
216             rate = done / el if el > 0 else 0.0
217             eta = (total - done) / rate if rate > 0 else 0.0
218             pct = 100 * done // total if total else 0
219             print(
220                 f"[fusion_golden]   {spec['name']}: detect frame {done}/{total} ({pct}%)
"
221                 f"| {el:.0f}s elapsed | ETA {eta:.0f}s",
222                 flush=True,

```



```

223         )
224         state["last"] = now
225
226     per_window = detector.detections_over_windows(
227         video_path, win_frames, frame_skip, progress_cb=progress
228     )
229
230     candidates: List[Dict[str, Any]] = []
231     for i, ((s, e), xs) in enumerate(zip(intervals, x_shuttle)):
232         det = per_window[i]
233         det_fw = det.get("frame_width") or fw
234         det_fh = det.get("frame_height") or 0.0
235         # Normalise the (pixel) net coord to 0-1 on the play axis, matching the foot-
point
236         # normalisation fusion_features uses (x/width, y/height).
237         if axis == "x":
238             net_norm = net_px / det_fw if det_fw > 0 else 0.5
239         else:
240             net_norm = net_px / det_fh if det_fh > 0 else 0.5
241         sf, ef = win_frames[i]
242         shuttle_in = [p for p in points if sf <= p.frame <= ef]
243         person = ff.compute_person_features(
244             det["frames"],
245             shuttle_in,
246             det_fw,
247             det_fh,
248             court_polygon=None,
249             net=(axis, net_norm),
250         )
251         candidates.append(
252             {
253                 "start": float(s),
254                 "end": float(e),
255                 "shuttle": {k: float(v) for k, v in zip(SHUTTLE_FEATURE_NAMES, xs)},
256                 "person": {k: float(v) for k, v in person.items()},
257                 "label": int(labels[i]),
258             }
259         )
260     return {
261         "name": spec["name"],
262         "fps": fps,
263         "frame_width": fw,
264         "candidates": candidates,
265         "gts": [[float(a), float(b)] for a, b in gts],
266     }
267
268
269 def run(
270     manifest_path: str,
271     out_dir: str,
272     frame_skip: int = 3,
273     iou: float = 0.5,
274     fresh: bool = False,
275     smallest_first: bool = False,
276     source_manifest: Optional[str] = None,
277     only: Sequence[str] = (),
278     missing_fusion_only: bool = False,
279     use_gpu: bool = True,
280     detector_model: Optional[str] = None,
281     max_gpu_temp_c: Optional[float] = None,
282 ) -> List[str]:
283     specs = load_resolved_manifest(manifest_path)
284     source_manifest = source_manifest or os.environ.get(SOURCE_MANIFEST_ENV)
285     if source_manifest:
286         updated = attach_source_video_paths(specs, source_manifest)

```

```

287         print(
288             f"[fusion_golden] attached {updated}/{len(specs)} source video path(s) "
289             f"from {source_manifest}",
290             flush=True,
291         )
292     specs = _filter_specs(specs, only)
293     if smallest_first:
294         # Process small videos first for quick feedback (the slow full match lands
295         # last);
296         # the LOVO result is order-independent. Proxy size by the trajectory CSV (one
297         # row/frame).
298         specs = sorted(
299             specs,
300             key=lambda s: (
301                 os.path.getsize(s["trajectory"])
302                 if os.path.isfile(s.get("trajectory", ""))
303                 else 0
304             ),
305         )
306     os.makedirs(out_dir, exist_ok=True)
307     # Run-telemetry black box: a long GPU run can be OS-killed with no traceback, so
308     # keep a tiny
309     # rotating runlog of velocity + machine health for the post-mortem. Constructing it
310     # must never
311     # break extraction, so guard the build and fall back to no telemetry on any error.
312     telem: Optional[RunTelemetry]
313     try:
314         telem = RunTelemetry(
315             os.path.join(out_dir, "run_telemetry.jsonl"), label="fusion_golden"
316         )
317     except Exception: # noqa: BLE001 - telemetry wiring must never break the run it
318         # observes
319         telem = None
320     written: List[str] = []
321     for vi, spec in enumerate(specs, 1):
322         out = os.path.join(out_dir, f"{spec['name']}_golden_features.json")
323         existing = os.path.isfile(out)
324         existing_full = existing and is_full_fusion_feature_file(out)
325         # Resumable: skip a video whose full-fusion JSON already exists. A leaner
326         # windowing-only JSON is stale for A7, so re-extract it even without --fresh.
327         if missing_fusion_only and not fresh and existing_full:
328             print(
329                 f"[fusion_golden] ({vi}/{len(specs)}) {spec['name']}: full-fusion "
330                 f"features already present -> {out}",
331                 flush=True,
332             )
333             written.append(out)
334             continue
335         if not fresh and existing and existing_full:
336             print(
337                 f"[fusion_golden] ({vi}/{len(specs)}) {spec['name']}: already extracted "
338                 f"-> {out} (use --fresh to redo)",
339                 flush=True,
340             )
341             written.append(out)
342             continue
343         if not fresh and existing and not existing_full:
344             print(
345                 f"[fusion_golden] ({vi}/{len(specs)}) {spec['name']}: existing JSON "
346                 f"is not full-fusion schema; re-extracting...",
347                 flush=True,
348             )
349         print(
350             f"[fusion_golden] ({vi}/{len(specs)}) {spec['name']}: extracting...",

```

```
346         flush=True,
347     )
348     t0 = time.monotonic()
349     if telem is not None:
350         telem.event("video_start", spec["name"])
```

8. assistant (2026-07-04T00:45:16.601Z)

Now let me check the ALPHA_LAUNCH_READINESS.md file more closely for context on the tracker states:

9. user (2026-07-04T00:45:17.423Z)

<system-reminder>Warning: the file exists but is shorter than the provided offset (545). The file has 267 lines.</system-reminder>

10. assistant (2026-07-04T00:45:19.216Z)

Let me read that tracker file completely:

11. user (2026-07-04T00:45:19.406Z)

```
1      # Alpha Launch Readiness Plan
2
3      > **Status:** `DRAFT` - cross-repo execution tracker; **Owner:** @avidullu; **Created:**
2026-07-03; **Last updated:** 2026-07-04
4      > **Lifecycle:** `DRAFT` -> IN PROGRESS -> DONE -> archived`
5      > **Tracking anchors:** Section 7 is the progress tracker and source of truth; indexed
in `docs/README.md` and `docs/DOC_STATUS.md`.
6      > **Relation to existing docs:** consolidates the launch-relevant pending items from
`NEXT_STEPS.md`, `ROADMAP.md`, `docs/data_pipeline/GOLDEN_VIDEOS.md`,
`R2_EVAL_METRIC_DESIGN.md`, `COMPUTE_DECOUPLED_SERVING/`, and the sibling `kheIsutra` Studio
docs.
7      > **Honesty note:** `[verified]` means checked against the local synced repos on
2026-07-03. `[analysis]` means derived from a local command run on that snapshot. This is not
legal, financial, or launch approval.
8
9      ---
10
11     ## 0. TL;DR
12
13     The corpus is now large enough to move from "grow labels first" into alpha-readiness
work, but the 15-video headline is not itself a launch gate pass. `[verified]` The golden
manifest has 15 badminton videos; `[analysis]` A2-A7 made the local manifest/path and full-
fusion/audio feature surface reproducible on all 15; `[analysis]` A4/A5 refreshed the n=15
nightly and heuristic floors; `[analysis]` A8/A9 still reject served Gate A default promotion
and owned-model serving claims on the current evidence.
14
15     The first controlled alpha should be a hosted, authenticated product loop: upload a
video, process it, select rallies, compile, and download a reel. The owned-model/commercial
rally-detection claim remains a separate NO-GO until the ship-gate target is ratified and the
WASB checkpoint provenance is resolved.
16
17     ## 1. Problem & goal
18
19     The previous launch blockers were spread across roadmap docs and mixed together three
different questions:
20
21     - **Can an invited tester use the product without SSH?**
22     - **Can the 15-video golden corpus now support honest quality gates?**
23     - **Can we claim or ship an owned/commercial rally detector?**
24
25     This doc separates those into executable workstreams. The goal is to reach an invite-
only alpha where the product is usable end-to-end, the quality claims are honest and
```

reproducible, and the remaining model/legal gates are explicit rather than accidentally implied by launch activity.

```

26
27     ## 2. Decisions locked
28
29     | # | Decision | Source / date | Implication |
30     |---|---|---|---|
31     | D1 | **Product alpha and owned-model launch are separate decisions.** |
`NEXT_STEPS.md` NO-GO banner, 2026-06-30; verified 2026-07-03 | We can launch a controlled
hosted product loop without claiming the owned model is commercially cleared. |
32     | D2 | **First alpha must be page-driven, not SSH/CLI-driven.** | `NEXT_STEPS.md` P0 UI-
driven/shareable item; verified 2026-07-03 | Hosted service, edge auth, upload, job polling,
compile, and download are launch-critical. |
33     | D3 | **The 15-video corpus is now a measurement asset, not a free launch pass.** |
`[verified]` `docs/data_pipeline/GOLDEN_VIDEOS.md`; `[analysis]` local eval runs, 2026-07-03 |
Rebase/portability, feature completeness, baseline refresh, and gate ratification come before
quality claims. |
34     | D4 | **Do not promote served Gate A by default from the current n=15 evidence.** |
`[analysis]` `backend.eval.serve_contrast` run, 2026-07-04 | Proposal-side lift does not survive
served-windowing safety; keep it as an opt-in/analysis lever until owner ratification. |
35     | D5 | **P10/P11 corpus promotion and train/re-eval are alpha-plus unless owner reopens
D13.** | sibling `khelesutra/docs/STUDIO_MEDIA_LIFECYCLE.md`; verified 2026-07-03 | First alpha
can use manual/offline corpus promotion; automated flywheel follows after serving and gates are
stable. |
36     | D6 | **Gen-0 n=15 shadow training is non-serving.** | `[analysis]`
`training.gen0.harness` run, 2026-07-04 | Learned mean beats the heuristic floor, but the paired
sign test is not significant; #172 and WASB C3 remain hard blockers. |
37
38     ## 3. Verified snapshot, 2026-07-03
39
40     All five Git checkouts were fast-forwarded before analysis:
41
42     | Repo | Branch | Synced HEAD |
43     |---|---|---|
44     | `badminton-highlight-indexer` | `master` | `e4914d1` |
45     | `khelesutra` | `master` | `b0f70a5` |
46     | `rally-annotator` | `main` | `ef9e088` |
47     | `sports-data-collector` | `master` | `bfc014f` |
48     | `sports-obsreport` | `main` | `a0c7252` |
49
50     Corpus/eval facts:
51
52     - `[verified]` `docs/data_pipeline/GOLDEN_VIDEOS.md` says
`output/human_lovo_manifest.json` is the ingested golden corpus at `n = 15`.
53     - `[analysis]` The local manifest had stale absolute paths. A temporary rebased manifest
found all 15 trajectories and labels in this workspace.
54     - `[analysis]` A7 generated full shuttle/person fusion schema for all 15 videos and A8
audio augmentation populated audio features for all 15.
55     - `[analysis]` `python -m backend.eval.fusion_compare --features-dir output` now runs
over all 15 and measures person-fusion `Delta F1 = +0.008`; the paired CI crosses zero, so
person fusion is not promotable by itself.
56     - `[analysis]` `python -m backend.eval.fusion_audio --features-dir output --compare`
measures audio `Delta F1 = +0.018`; the paired CI crosses zero, so audio is a shadow/suggestive
cue, not a default-flip.
57     - `[analysis]` `python -m backend.eval.serve_contrast --manifest
output\human_lovo_manifest.json` measured proposal Gate A mean `Delta F1 +0.0287`, but served
mean `Delta F1 -0.0328`; not served-safe.
58     - `[analysis]` `python -m training.gen0.harness --manifest
output\human_lovo_manifest.json --floor eval_baselines\heuristic_lovo_n15.json --version
0.1.1-n15-shadow.20260704 --out artifacts\rally_tal_gen0` produced learned LOVO mean F1 `0.551`,
floor passed versus heuristic `0.446`, but sign test `9/15`, `p = 0.3036`, `ship=False`.
59     - `[analysis]` `python scripts/nightly_regression.py --manifest <rebased-n15>
--features-dir output --no-report` reported default/milestone tolF1 `0.3636` vs baseline-off
tolF1 `0.1382`, but marked the frozen baseline stale due to config/corpus changes.
60     - `[analysis]` Before A6, `python -m backend.eval.ablation --features-dir output

```

```

--granularity group` failed at import due to a circular import between
`backend/eval/ablation.py` and `backend/eval/ablation_pdf.py`; A6 fixes the module entrypoint
and keeps full-fusion ablation scoped to the 6 videos with full schema until A7.
61
62     ## 4. Launch strategy
63
64     ### 4.1 Alpha lane: make the product usable
65
66     The alpha lane is successful when an invited tester can open the app, authenticate,
upload a video, watch the job progress, pick rallies, compile a reel, and download it. This lane
should avoid model-quality promises beyond the measured state and should default to the most
reliable serving path.
67
68     ### 4.2 Quality lane: make the 15-video corpus gate-ready
69
70     The quality lane is successful when the 15-video corpus is portable, reproducible,
feature-complete, and backed by refreshed baselines. Only then should R2 tolF1,
Gen-0/TemporalMaxer, or other model/promotion decisions be ratified.
71
72     ### 4.3 Flywheel lane: make tester feedback useful
73
74     The flywheel lane is successful when user corrections can flow into the collector/vault
path with consent/provenance and a visible re-eval delta. This is valuable for alpha learning,
but it should not block the first controlled alpha if manual promotion can bridge the gap.
75
76     ## 5. Risks and guardrails
77
78     | Risk | Why it matters | Guardrail |
79     |---|---|---|
80     | Launch activity implies model launch approval | The engine docs explicitly say rally-
detection product launch is NO-GO until ship target and WASB C3 clear | Keep all alpha copy
scoped to invite-only processing, not owned-model claims |
81     | Stale local manifests create false confidence | Manifest-driven tools skipped or
failed until paths were rebased | Make the corpus manifest portable and add a path/fixture smoke
check |
82     | 15-video corpus is only partially feature-complete | Fusion/person/audio eval is still
n=6 today | Regenerate or upconvert full-fusion features for all 15 before fusion decisions |
83     | Baseline refresh accidentally rubber-stamps a regression | Nightly runner correctly
reported stale baselines | Refresh baselines only after reviewer sign-off and record the
command/output |
84     | P10/P11 expands scope before hosted alpha is usable | Corpus flywheel is valuable but
can consume the alpha launch window | Keep P10/P11 behind explicit owner reopen unless alpha
lane is already stable |
85     | WASB checkpoint provenance remains unresolved | Blocks commercial sharing of a trained
model regardless of F1 | Treat C3 as a hard model-release gate; do not hide it inside product
launch tasks |
86
87     ## 6. Honest limits
88
89     Completing this doc's alpha lane does **not** prove the owned model is ready, that WASB
weights are commercially clean, that the product can serve unbounded public traffic, that multi-
sport is ready, or that every tester correction automatically improves the model.
90
91     Completing the quality lane does **not** by itself deploy a service. It only makes the
measurement surface honest enough for a launch decision.
92
93     ## 7. Deliverables & progress tracker
94
95     Legend: TODO = not started, IN PROGRESS = actively being worked, DONE =
shipped/verified, GATED = blocked on owner/legal/platform decision. One PR per row unless the
row explicitly names a cross-repo batch.
96
97     | ID | Deliverable | Repo / owner | Depends on | Gated? | Status | PR |
98     |---|-----|-----|-----|-----|-----|---|
99     | A0 | Create this alpha readiness tracker and index it in docs | `badminton-highlight-

```

```

indexer` | - | No | DONE | #451 |
100 | A1 | Stand up alpha serving endpoint: Cloud Run/GPU or approved host, health check,
edge auth ON, upload -> process -> jobs -> compile -> download verified | `khelsutra` +
`badminton-highlight-indexer` | A0 | Owner spend / CF config | TODO | - |
101 | A2 | Make the n=15 golden manifest portable/reproducible; add a path/corpus smoke
check that fails loudly when labels or trajectories are missing | `badminton-highlight-indexer`
| A0 | No | DONE | #452 |
102 | A3 | Promote the 15-video source-video/MD5 records into the collector/vault path;
reconcile collector's six-video source manifest with the 15-video eval corpus | `sports-data-
collector` + vault | A2 | Owner upload/storage scope | IN PROGRESS | collector #62 |
103 | A4 | Refresh the n=15 nightly regression baseline intentionally and record the exact
command/output; keep stale-baseline warnings until reviewed | `badminton-highlight-indexer` | A2
| Reviewer sign-off | DONE | #453 |
104 | A5 | Recompute the heuristic n=15 floor and replace the stale `heuristic_lovo_n6`
floor for future Gen-0 comparisons | `badminton-highlight-indexer` | A2 | No | DONE | #454 |
105 | A6 | Fix `backend.eval.ablation` CLI circular import, then run the R2/tolF1 ablation
on the n=15 corpus | `badminton-highlight-indexer` | A2 | No | DONE | #455 |
106 | A7 | Regenerate or upconvert full-fusion feature JSONs for the newer 9 videos so
fusion/person/audio analyses run over all 15 | `badminton-highlight-indexer` | A2 | GPU/data
availability | IN PROGRESS | this PR |
107 | A8 | Re-run `fusion_compare`, group ablation, and served contrast after A7; explicitly
promote/reject person fusion, Gate A served-default, and any other R-series default flips |
`badminton-highlight-indexer` | A6, A7 | Reviewer sign-off | IN PROGRESS | this PR |
108 | A9 | Run Gen-0/TemporalMaxer shadow training against the refreshed n=15 floor; produce
a non-serving artifact bundle and gate verdict | `badminton-highlight-indexer` | A5, A8 | Owner
approval for M0.4 scope | IN PROGRESS | this PR |
109 | A10 | Decide and record the alpha ship-gate target: metric, corpus slice, minimum
threshold, significance rule, and what claims are allowed | Owner + `badminton-highlight-
indexer` | A4, A5, A8 | Owner decision | GATED | - |
110 | A11 | Resolve or explicitly defer WASB C3 provenance for alpha messaging; if deferred,
document "no commercial model sharing" copy guardrail | Owner/legal | A10 | Legal/provenance |
GATED | - |
111 | A12 | Reopen or defer P10/P11 corpus flywheel: `PATCH /api/segments`, collector
promotion, `train`/`reeval` job, delta-F1 surface, and alignment with
`COMPUTE_DECOUPLED_SERVING` M11 | `khelsutra` + `badminton-highlight-indexer` + collector | A1,
A3 | Owner D13 reopen (`khelsutra/docs/STUDIO_MEDIA_LIFECYCLE.md`) | GATED | - |
112 | A13 | Alpha launch review: runbook, privacy/consent copy, observability/job-failure
triage, known limits, rollback/disable steps, and owner GO/NO-GO | Cross-repo | A1, A4, A10, A11
| Owner launch decision | TODO | - |
113
114 ## 8. Concrete next PR sequence
115
116 1. **Docs PR:** land this tracker.
117 2. **Corpus portability PR:** rebase/portable manifest tooling plus missing-file smoke
test.
118 3. **Hosted alpha PR/batch:** health/auth/upload/process/job/compile/download
verification and runbook updates.
119 4. **Baseline PR:** intentional n=15 nightly baseline refresh and heuristic floor
refresh.
120 5. **Ablation/fusion PRs:** fix ablation import, complete 15-video features, rerun
promotion/rejection decisions.
121 6. **Launch decision PR/docs update:** record ship-gate target, C3 status, alpha copy
limits, and launch checklist.
122
123 A2 smoke command:
124
125 ```powershell
126 python -m backend.eval.golden_manifest --manifest output/human_lovo_manifest.json --min-
count 15
127 python -m backend.eval.golden_manifest --manifest output/human_lovo_manifest.json --min-
count 15 --write output/human_lovo_manifest.local.json
128 ```
129
130 The checker resolves stale absolute paths against the repo's sibling `Annotation Setup`
folders and `output/`, then exits non-zero if any required `trajectory` or `labels` file is

```

```

absent.
131
132     A4 nightly baseline refresh record:
133
134     ```powershell
135     python scripts\nightly_regression.py --manifest output\human_lovo_manifest.json
--features-dir output --update-baseline
136     ```
137
138     Result:
139
140     ```text
141     INFO baseline snapshot written to C:\Users\avidu\Projects\khelsutra-guru\badminton-
highlight-indexer\eval_baselines\nightly_baseline.json
142     INFO default: n=15 tolF1=0.364 f1@0.5=0.472 merge_rate=0.135
143     INFO milestone: n=15 tolF1=0.364 f1@0.5=0.472 merge_rate=0.135
144     INFO baseline_off: n=15 tolF1=0.138 f1@0.5=0.237 merge_rate=0.638
145     ```
146
147     Pass-mode proof:
148
149     ```powershell
150     python scripts\nightly_regression.py --manifest output\human_lovo_manifest.json
--features-dir output --no-report
151     ```
152
153     Result: `PASS`, exit code `0`, baseline snapshot `2026-07-03`, default/milestone `tol_f1
= 0.3636`, `f1@0.5 = 0.4723`, `merge_rate = 0.1350`; baseline-off `tol_f1 = 0.1382`, `f1@0.5 =
0.2366`, `merge_rate = 0.6380`.
154
155     A5 heuristic floor refresh record:
156
157     ```powershell
158     python -m backend.eval.calibrate_local --manifest output\human_lovo_manifest.json
--metric f1 --floor-out eval_baselines\heuristic_lovo_n15.json
159     ```
160
161     Result:
162
163     ```text
164     baseline (untuned) mean held-out f1: 0.237
165     calibrated mean held-out f1:          0.446 +/- 0.239 (gap +0.000)
166     ```
167
168     The committed floor is `eval_baselines/heuristic_lovo_n15.json`: `n = 15`, `mean_f1 =
0.4456`, `std = 0.2390`, `metric = f1`, `iou = 0.5`. Every leave-one-video-out fold selected
`(0.05, 1.0, 2.0, 0)`.
169
170     A6 ablation/R2 record:
171
172     ```powershell
173     python -m backend.eval.ablation --features-dir output --granularity group
174     python -m backend.eval.rally_seg_eval --manifest output\human_lovo_manifest.json
--features-dir output --preset served --max-window-duration 0 --inactivity-end 0 --vs served
--vs-max-window-duration 6 --vs-inactivity-end 1.5 --json-out output\a6_r2_ablation.json
175     ```
176
177     Result: the ablation CLI now runs, but full-fusion feature ablation still reports `n=6`
because only the original six feature JSONs carry the full fusion schema; A7 must
regenerate/upconvert the newer nine before fusion/person/audio promotion decisions. The n=15
R2/tolF1 contrast is positive for the served milestone versus baseline-off: baseline/off `tolF1
= 0.138`, served `tolF1 = 0.364`, `Delta tolF1 = +0.225` with CI `[+0.165, +0.283]`, sign
`14/0`, `p = 0.000`; `Delta F1@0.5 = +0.236` with CI `[+0.161, +0.313]`; merge rate improves
from `0.638` to `0.135`. Net over-segmentation guard passes (`1.276 -> 0.378`), while the strict
segment-count rule remains blocked by increased splits, so R2/tolF1 is the supported gate for

```

the baseline-off to served milestone contrast rather than a blanket fusion/default-flip approval.

178

179 A7 full-fusion/audio feature refresh record:

180

181 ```powershell

182 python -m backend.eval.fusion_golden --manifest output\human_lovo_manifest.json --out-dir output --source-manifest ..\sports-data-collector\data\golden_source_videos.json --missing-fusion-only --smallest-first --max-gpu-temp-c 87

183 python -m backend.eval.fusion_audio --manifest output\human_lovo_manifest.json --features-dir output --source-manifest ..\sports-data-collector\data\golden_source_videos.json --augment

184 ```

185

186 Result: the GPU pass completed without a thermal abort after fans were set to max performance. Full-fusion schema is now present for `15/15` feature JSONs, and audio features are present for `15/15`. The large regenerated clips completed as follows: `mahadevpura\_1` `1` window / `0` positives / `10` GT, `GX020094` `99` / `30` / `40`, `GX030094` `79` / `21` / `41`, `GX010141` `26` / `11` / `29`, `GX010137` `56` / `31` / `34`, `Boxhill\_Doubles` `59` / `12` / `43`, `mahadevpura\_2` `14` / `2` / `40`, `GX010128` `51` / `15` / `52`, `Badminton\_BXH\_2` `29` / `1` / `44`. The source manifest supplied all 15 source-video paths, including the F-drive Boxhill originals.

187

188 A8 fusion/default-decision record:

189

190 ```powershell

191 python -m backend.eval.fusion_compare --features-dir output

192 python -m backend.eval.fusion_audio --features-dir output --compare

193 python -m backend.eval.ablation --features-dir output --granularity group

194 python -m backend.eval.serve_contrast --manifest output\human_lovo_manifest.json

195 python -m backend.eval.rally_seg_eval --manifest output\human_lovo_manifest.json --features-dir output --preset served --max-window-duration 0 --inactivity-end 0 --vs served --vs-max-window-duration 6 --vs-inactivity-end 1.5 --json-out output\A6_r2_ablation.json

196 ```

197

198 Result:

199

200 - Person fusion is **not promoted**: shuttle-only F1 `0.302`, shuttle+person F1 `0.311`, `Delta F1 +0.008`, paired CI `[-0.040, +0.055]`, sign `8W/6L`, `p = 0.791`.

201 - Audio is **shadow/suggestive, not promoted**: shuttle+person F1 `0.311`, shuttle+person+audio F1 `0.329`, `Delta F1 +0.018`, paired CI `[-0.017, +0.053]`, sign `10W/4L`, `p = 0.180`.

202 - Group ablation remains non-significant for cue promotion: audio `+0.018` CI `[-0.017, +0.053]`; shuttle `+0.013` CI `[-0.002, +0.029]`; gate_a `-0.001` CI `[-0.012, +0.008]`; person `-0.009` CI `[-0.045, +0.021]`.

203 - Gate A served default is **rejected**: proposal mean `Delta F1 +0.0287`, CI `[-0.0222, +0.0680]`; served mean `Delta F1 -0.0328`, CI `[-0.0799, +0.0052]`, `regresses=True`.

204 - R2/cap/inactivity remains the strong quality result: served candidate F1 `0.472` vs baseline-off `0.237`, `Delta F1 +0.236`, CI `[+0.161, +0.313]`; tolF1 `0.364` vs `0.138`, `Delta tolF1 +0.225`, CI `[+0.165, +0.283]`; merge rate `0.135` vs `0.638`. Net over-seg guard passes, strict per-video split-count guard still blocks blanket promotion language.

205

206 A9 Gen-0 shadow-training record:

207

208 ```powershell

209 python -m training.gen0.harness --manifest output\human_lovo_manifest.json --floor eval_baselines\heuristic_lovo_n15.json --out artifacts\rally_tal_gen0 --version 0.1.1-n15-shadow.20260704

210 ```

211

212 Result: current Gen-0 LogReg/TAL harness produced local bundle `artifacts\rally\_tal\_gen0\0.1.1-n15-shadow.20260704` with learned LOVO mean F1 `0.551` versus heuristic floor `0.4456`, so `floor\_passed=True`. The gate remains **non-serving**: paired sign test `9/15`, `p = 0.3036`, `ship=False`; blockers are uncalibrated ship-gate target #172 and WASB checkpoint provenance C3. TemporalMaxer itself is not implemented in this branch; the

evidence supports either closing A9 as a Gen-0 shadow baseline or opening a follow-up M0.4 scorer-swap PR.

213

214 ## 9. Open questions

215

216 1. ****Owner:**** Is the first alpha allowed to use manual/offline corpus promotion, or should P10/P11 be reopened before any tester wave?

217 2. ****Owner/platform:**** Which hosted compute target is the next stand-up: Cloud Run GPU, GCE L4 VM, or another approved GPU host?

218 3. ****Owner/legal:**** What is the acceptable alpha wording while WASB C3 remains unresolved?

219 4. ****Engineering:**** Should the portable corpus manifest live as a generated artifact, a checked-in redacted manifest, or both?

220 5. ****Engineering:**** Should A9 close on the current Gen-0 non-serving bundle, or should the actual TemporalMaxer scorer swap be implemented before marking A9 done?

221

222 ## 10. Definition of done

223

224 - [] Invited tester can complete the hosted record/upload -> reel download CUJ without SSH.

225 - [] Alpha service is gated behind the chosen auth posture and has a documented disable/rollback path.

226 - [x] n=15 corpus is portable enough that manifest-driven quality commands do not rely on stale local absolute paths.

227 - [x] n=15 nightly baseline and heuristic floor are refreshed with recorded commands.

228 - [x] R2/tolF1 gate is promoted or rejected with a recorded ablation result.

229 - [x] Fusion/person/Gate-A served-default decisions are explicitly promoted or rejected on the 15-video corpus.

230 - [] Ship-gate target and allowed alpha claims are recorded.

231 - [] WASB C3 status is recorded and reflected in alpha messaging.

232 - [] P10/P11 are either reopened with a committed sequence or explicitly deferred for alpha-plus.

233 - [] Owner records GO/NO-GO for the first alpha wave.

234

235 ## 11. References

236

237 Internal docs:

238

239 - `docs/NEXT\_STEPS.md`

240 - `docs/ROADMAP.md`

241 - `docs/data\_pipeline/GOLDEN\_VIDEOS.md`

242 - `docs/R2\_EVAL\_METRIC\_DESIGN.md`

243 - `docs/RALLY\_DETECTION\_FIXES\_PLAN.md`

244 - `docs/COMPUTE\_DECOUPLED\_SERVING/README.md`

245 - sibling `kheksutra/NEXT\_STEPS.md`

246 - sibling `kheksutra/docs/STUDIO\_MEDIA\_LIFECYCLE.md`

247 - sibling `kheksutra/docs/PER\_RALLY\_SELECTION.md`

248 - sibling `sports-data-collector/data/golden\_source\_videos.json`

249

250 Local verification commands used for the 2026-07-03 snapshot:

251

252 ```powershell

253 python -m backend.eval.fusion_compare --features-dir output

254 python -m backend.eval.serve_contrast --manifest <rebased-n15-manifest>

255 python -m training.gen0.harness --manifest <rebased-n15-manifest> --floor eval_baselines\heuristic_lovo_n15.json --version analysis-n15 --out <temp-out>

256 python scripts\nightly_regression.py --manifest <rebased-n15-manifest> --features-dir output --no-report

257 python -m backend.eval.ablation --features-dir output --granularity group

258 python -m backend.eval.rally_seg_eval --manifest output\human_lovo_manifest.json --features-dir output --preset served --max-window-duration 0 --inactivity-end 0 --vs served --vs-max-window-duration 6 --vs-inactivity-end 1.5 --json-out output\A6_R2_ablation.json

259 ```

260

261 ---

```
262
263     ### Changelog
264
265     - 2026-07-03 - Created from a synced five-repo review of `NEXT_STEPS`, `ROADMAP`,
golden-corpus docs, local eval artifacts, and the current 15-video corpus state.
266     - 2026-07-04 - Recorded A7 full-fusion/audio regeneration, A8 n15 fusion/default
decisions, and A9 Gen-0 n15 shadow-training gate verdict.
267
```

12. assistant (2026-07-04T00:45:21.556Z)

Now let me check the training.gen0.harness module to understand the A9 implementation:

13. user (2026-07-04T00:45:22.576Z)

find: ?C:\Users\avidu\Projects\khelsutra-gurubadminton-highlight-indexer?: No such file or directory

14. user (2026-07-04T00:45:32.584Z)

training\gen0\harness.py

15. user (2026-07-04T00:45:34.602Z)

```
1     """Gen-0 TAL harness (M0.4): ONE command -> train -> honest LOVO eval -> ship-gate ->
artifact.
2
3     python -m training.gen0.harness --manifest output/human_lovo_manifest.json \
4         --floor eval_baselines/heuristic_lovo_n15.json --out artifacts/rally_tal_gen0
--version 0.1.0
5
6     Productionizes the rally_seq prototype per ADR-008:
7     - TRAINS on the corpus manifest (per-video trajectory + golden labels), evaluates with
8       honest leave-one-video-out (split by match, threshold tuned on train only), then fits
9       the FINAL model on all videos.
10    - Emits a versioned ARTIFACT BUNDLE ? the only thing serving may ever consume:
11        <out>/<version>/weights.json    (gitignored; classifier + frozen normalization)
12        <out>/<version>/manifest.json    (the reviewable contract: feature-schema version,
13                                          calibration, training lineage + GT hashes,
14                                          eval results, attestations, gate verdict)
15    - Runs the SHIP-GATE as pre-flight: floor comparison + a PAIRED per-video sign test
16      (at n=6 a mean gap alone is noise-ambiguous), plus the hard attestation blocks
17      (WASB C3 unresolved => commercial ship refused; explicit F1@tIoU target still
18      undefined => no self-declared victory). The product side owns final admission
19      (backend/eval/regression.py); this harness never grades itself into serving.
20
21    Training may import backend.* (one-way wall); backend must never import training.*.
22    """
23
24    from __future__ import annotations
25
26    import argparse
27    import hashlib
28    import json
29    import os
30    import re
31    import subprocess
32    import time
33    from math import comb
34    from typing import Dict, List, Optional
35
36    import numpy as np
37
38    from backend.eval.classifier import LogisticRegression
39    from backend.eval.golden_manifest import load_resolved_manifest
```

```

40     from backend.eval.gt_loader import load_gt_intervals
41     from backend.eval.rally_seq_proto import (
42         _seg_f1,
43         labels_for,
44     )
45     from backend.eval.rally_seq_proto import (
46         run as lovo_run,
47     )
48     from backend.eval.regression import gt_hash
49     from backend.features.rally_seq import (
50         FEAT_NAMES,
51         FEATURE_SCHEMA,
52         build_frame_arrays,
53         features,
54     )
55     from backend.pipeline.detectors.tracknet_runner import TrackNetRunner
56
57     # Post-processing constants ? part of the model's identity, recorded in the manifest.
58     CALIBRATION_DEFAULTS = {"smooth": 7, "merge_gap": 1.0, "min_dur": 1.0}
59     THRESHOLD_GRID = [round(t, 2) for t in np.arange(0.2, 0.81, 0.05)]
60
61
62     # Repo root = two levels up from training/gen0/harness.py. A `GIT_SHA` file baked here
63     at
64     # tarball-build time is the off-box provenance fallback (see _git_sha +
65     deploy/gcp/README.md).
66     _REPO_ROOT = os.path.abspath(os.path.join(os.path.dirname(__file__), "..", ".."))
67     _SHA_RE = re.compile(r"^[0-9a-f]{7,40}$")
68
69     #: Env var carrying the source commit SHA out-of-band when the run has no `.git` (a `git
70     archive`
71     #: tarball ? the GCP-L4 path). Set it on the box, or bake a `GIT_SHA` file into the
72     archive root.
73     GIT_SHA_ENV = "RALLY_GIT_SHA"
74
75     def _valid_sha(s: Optional[str]) -> Optional[str]:
76         """Return the normalized commit SHA if `s` looks like one (7-40 hex chars), else
77         None.
78
79         Guards every provenance source so a stray value (`unknown\n`, a build label) can
80         never be
81         stamped as if it were a real SHA."""
82         s = (s or "").strip().lower()
83         return s if _SHA_RE.match(s) else None
84
85     def _git_sha() -> str:
86         """Resolve the source commit SHA for the manifest, in priority order:
87
88         1. live ``git rev-parse HEAD`` ? a real checkout (local dev / CI); authoritative,
89         unchanged.
90         2. the ``RALLY_GIT_SHA`` env var ? out-of-band injection for a run off a ``git
91         archive``
92         tarball, which carries no ``.git`` (the 2026-06-20 GCP-L4 box stamped ``unknown``
93         here).
94         3. a ``GIT_SHA`` file baked into the repo root at tarball-build time.
95         4. ``"unknown"`` ? nothing available (last resort; never fabricated).
96
97         Steps 2-4 make a remote tarball run provenance-stamped WITHOUT changing the git-
98         present path.
99         """
100         try:
101             out = subprocess.run(
102                 ["git", "rev-parse", "HEAD"], capture_output=True, text=True, timeout=10

```

```

95         ).stdout
96         live = _valid_sha(out)
97         if live:
98             return live
99     except Exception:
100         pass
101     env_sha = _valid_sha(os.environ.get(GIT_SHA_ENV))
102     if env_sha:
103         return env_sha
104     try:
105         with open(os.path.join(_REPO_ROOT, "GIT_SHA"), encoding="utf-8") as f:
106             file_sha = _valid_sha(f.read())
107             if file_sha:
108                 return file_sha
109     except OSError:
110         pass
111     return "unknown"
112
113
114 def _sha256(path: str) -> str:
115     h = hashlib.sha256()
116     with open(path, "rb") as f:
117         for chunk in iter(lambda: f.read(65536), b''):
118             h.update(chunk)
119     return h.hexdigest()
120
121
122 def sign_test(learned: List[float], floor: List[float]):
123     """Paired one-sided sign test: is learned > floor more often than chance?
124
125     At n=6 a mean comparison alone is inside noise (SE ~0.11); the paired per-video
126     wins are the defensible statistic. Returns (wins, n, p_one_sided)."""
127     pairs = [
128         (lv, fv) for lv, fv in zip(learned, floor) if lv != fv
129     ] # ties drop, standard
130     wins = sum(1 for lv, fv in pairs if lv > fv)
131     n = len(pairs)
132     p = sum(comb(n, k) for k in range(wins, n + 1)) / (2**n) if n else 1.0
133     return wins, n, p
134
135
136 def train_final(specs: List[Dict]):
137     """Fit the FINAL model on ALL corpus videos; threshold tuned on the same (train-
138     side).
139
140     The honest generalization estimate is the LOVO result, not this fit ? recorded as
141     such."""
142     vids = []
143     for s in specs:
144         pts = TrackNetRunner.parse_trajectory_csv(s["trajectory"])
145         arr = build_frame_arrays(pts, float(s["frame_width"]), float(s["fps"]))
146         X, times = features(arr, float(s["fps"]))
147         gts = load_gt_intervals(s["labels"], strict=False)
148         vids.append(
149             {
150                 "name": s["name"],
151                 "X": X,
152                 "times": times,

```

16. assistant (2026-07-04T00:45:36.709Z)

Let me search for TemporalMaxer references:

17. user (2026-07-04T00:45:37.749Z)

No files found

18. assistant (2026-07-04T00:45:40.680Z)

Good ? so TemporalMaxer is not implemented. Now let me examine the golden_manifest.py to understand path resolution:

19. user (2026-07-04T00:45:40.985Z)

backend\eval\golden_manifest.py

20. user (2026-07-04T00:45:43.122Z)

```
1      """Portable golden-corpus manifest loader and smoke checker.
2
3      The owner-side golden manifest is intentionally gitignored and has historically
4      carried absolute paths from whichever box ingested a clip. This module keeps that
5      artifact usable after the workspace moves by resolving stale paths against the
6      current repo layout and failing loudly when required corpus files are absent.
7
8      Default local layout supported:
9      - ../Annotation Setup/Trajectories/<trajectory>.csv
10     - ../Annotation Setup/Golden Labelled/<labels>.csv
11     - output/<labels>.csv
12
13     Extra roots can be supplied with ``--root`` or ``RALLY_GOLDEN_SEARCH_ROOTS``.
14     """
15
16     from __future__ import annotations
17
18     import argparse
19     import json
20     import os
21     import sys
22     from dataclasses import dataclass
23     from pathlib import Path, PureWindowsPath
24     from typing import Any, Iterable, Sequence
25
26     DEFAULT_MANIFEST = "output/human_lovo_manifest.json"
27     SEARCH_ROOTS_ENV = "RALLY_GOLDEN_SEARCH_ROOTS"
28     DEFAULT_REQUIRED_FIELDS = ("trajectory", "labels")
29
30     _FIELD_DIR = {
31         "trajectory": "Trajectories",
32         "labels": "Golden Labelled",
33     }
34
35
36     @dataclass(frozen=True)
37     class ManifestIssue:
38         """One manifest path problem found by the resolver."""
39
40         name: str
41         field: str
42         path: str
43         reason: str
44
45         def message(self) -> str:
46             shown = f" {self.path!r}" if self.path else ""
47             return f"{self.name}: {self.field}{shown} - {self.reason}"
48
49
50     @dataclass(frozen=True)
51     class ManifestRebase:
```

```

52     """One path field changed from a stale manifest value to a local file."""
53
54     name: str
55     field: str
56     old_path: str
57     new_path: str
58
59     def message(self) -> str:
60         return f"{self.name}: {self.field} {self.old_path!r} -> {self.new_path!r}"
61
62
63     class ManifestPathError(RuntimeError):
64         """Raised when a strict manifest load finds missing required files."""
65
66         def __init__(self, manifest_path: str | os.PathLike[str], issues:
Sequence[ManifestIssue]):
67             self.manifest_path = os.fspath(manifest_path)
68             self.issues = list(issues)
69             detail = "\n".join(f" - {i.message()}" for i in self.issues[:12])
70             more = "" if len(self.issues) <= 12 else f"\n ... {len(self.issues) - 12} more"
71             super().__init__(
72                 f"golden manifest path check failed for {self.manifest_path}: "
73                 f"{len(self.issues)} issue(s)\n{detail}{more}"
74             )
75
76
77     def repo_root() -> Path:
78         """Return the repository root for this module."""
79
80         return Path(__file__).resolve().parents[2]
81
82
83     def load_manifest(manifest_path: str | os.PathLike[str]) -> list[dict[str, Any]]:
84         """Load the flat golden manifest list and validate its outer shape."""
85
86         path = Path(manifest_path)
87         with path.open(encoding="utf-8") as f:
88             data = json.load(f)
89             if not isinstance(data, list):
90                 raise ValueError(f"golden manifest must be a JSON list: {path}")
91             out: list[dict[str, Any]] = []
92             for i, row in enumerate(data):
93                 if not isinstance(row, dict):
94                     raise ValueError(f"golden manifest row {i} is not an object: {path}")
95                 out.append(dict(row))
96             return out
97
98
99     def resolve_manifest_paths(
100         manifest_path: str | os.PathLike[str],
101         *,
102         repo_root_path: str | os.PathLike[str] | None = None,
103         extra_roots: Iterable[str | os.PathLike[str]] = (),
104         required_fields: Sequence[str] = DEFAULT_REQUIRED_FIELDS,
105     ) -> tuple[list[dict[str, Any]], list[ManifestIssue], list[ManifestRebase]]:
106         """Load and rebase required path fields in a golden manifest.
107
108         Returns ``(resolved_specs, issues, rebased_paths)``. The returned specs preserve
109         all manifest metadata but replace any resolved ``trajectory`` / ``labels`` path
110         with the existing local path. Missing or ambiguous required fields are reported
111         in ``issues``.
112         """
113
114         manifest = load_manifest(manifest_path)
115         return resolve_manifest_specs(

```

```

116         manifest,
117         manifest_path,
118         repo_root_path=repo_root_path,
119         extra_roots=extra_roots,
120         required_fields=required_fields,
121     )
122
123
124 def load_resolved_manifest(
125     manifest_path: str | os.PathLike[str],
126     *,
127     repo_root_path: str | os.PathLike[str] | None = None,
128     extra_roots: Iterable[str | os.PathLike[str]] = (),
129     required_fields: Sequence[str] = DEFAULT_REQUIRED_FIELDS,
130     strict: bool = True,
131 ) -> list[dict[str, Any]]:
132     """Load a manifest with stale paths rebased; optionally raise on missing files."""
133
134     specs, issues, _rebases = resolve_manifest_paths(
135         manifest_path,
136         repo_root_path=repo_root_path,
137         extra_roots=extra_roots,
138         required_fields=required_fields,
139     )
140     if strict and issues:
141         raise ManifestPathError(manifest_path, issues)
142     return specs
143
144
145 def resolve_manifest_specs(
146     specs: Sequence[dict[str, Any]],
147     manifest_path: str | os.PathLike[str],
148     *,
149     repo_root_path: str | os.PathLike[str] | None = None,
150     extra_roots: Iterable[str | os.PathLike[str]] = (),
151     required_fields: Sequence[str] = DEFAULT_REQUIRED_FIELDS,
152 ) -> tuple[list[dict[str, Any]], list[ManifestIssue], list[ManifestRebase]]:
153     """Resolve an already-loaded manifest list."""
154
155     manifest = Path(manifest_path)
156     root = Path(repo_root_path).resolve() if repo_root_path else repo_root()
157     roots = [Path(p) for p in _env_roots()]
158     roots.extend(Path(p) for p in extra_roots)
159     required = tuple(required_fields)
160
161     resolved: list[dict[str, Any]] = []
162     issues: list[ManifestIssue] = []
163     rebases: list[ManifestRebase] = []
164     for idx, spec in enumerate(specs):
165         row = dict(spec)
166         name = str(row.get("name") or f"row[{idx}]")
167         for field in required:
168             raw = row.get(field)
169             if not raw:
170                 issues.append(ManifestIssue(name, field, "", "required field missing"))
171                 continue
172             matches = resolve_existing_paths(
173                 str(raw),
174                 field,
175                 manifest_path=manifest,
176                 repo_root_path=root,
177                 extra_roots=roots,
178             )
179             if len(matches) > 1:
180                 found = "; ".join(str(p) for p in matches)

```

```

181             issues.append(
182                 ManifestIssue(
183                     name,
184                     field,
185                     str(raw),
186                     f"ambiguous file matches for basename: {found}",
187                 )
188             )
189         elif not matches:
190             issues.append(ManifestIssue(name, field, str(raw), "file not found"))
191         else:
192             hit = matches[0]
193             row[field] = str(hit)
194             if not _same_path(str(raw), hit):
195                 rebases.append(ManifestRebase(name, field, str(raw), str(hit)))
196             resolved.append(row)
197         return resolved, issues, rebases
198
199
200     def resolve_existing_paths(

```

21. assistant (2026-07-04T00:45:45.692Z)

Perfect. Now let me check the structure of what's happening with path resolution across the codebase:

22. user (2026-07-04T00:45:46.968Z)

355dad6 eval: refresh n15 fusion and shadow training

23. assistant (2026-07-04T00:45:49.530Z)

Now let me examine if there are any references to how the schema versioning or explicit tags for JSON structure:

24. user (2026-07-04T00:45:50.550Z)

```

C:\Users\avidu\Projects\khehsutra-guru\badminton-highlight-
indexer\backend\eval\ablation.py-1-""Leave-one-signal-out / leave-one-group-out ablation runner
(Phase 0).
C:\Users\avidu\Projects\khehsutra-guru\badminton-highlight-indexer\backend\eval\ablation.py-2-
backend\eval\ablation.py:3:A *systematic, repeatable* version of the one-off ablation
(``scratch/gate_ab_ablation.py``):
C:\Users\avidu\Projects\khehsutra-guru\badminton-highlight-
indexer\backend\eval\ablation.py-4-run it and it EXCLUDES a single signal, measures whether
quality dropped, and reports ? per
C:\Users\avidu\Projects\khehsutra-guru\badminton-highlight-
indexer\backend\eval\ablation.py-5-signal ? the **marginal contribution** ?F1 (full set vs full-
minus-that-signal) with a bootstrap
--
C:\Users\avidu\Projects\khehsutra-guru\badminton-highlight-indexer\backend\eval\ablation.py-457-
the loader never crashes, mirroring ``experiment._feature_column``'s tolerance.
C:\Users\avidu\Projects\khehsutra-guru\badminton-highlight-indexer\backend\eval\ablation.py-458-
backend\eval\ablation.py:459:   Leaner velocity-windowing-only files (no ``shuttle````person``
cue dicts ? the schema the
C:\Users\avidu\Projects\khehsutra-guru\badminton-highlight-indexer\backend\eval\ablation.py-460-
R3/R4 corpus-growth path persists) are SKIPPED, not crashed on, so a mixed corpus loads the
backend\eval\ablation.py:461:   ablatable subset cleanly (see ``_is_fusion_schema``).""
C:\Users\avidu\Projects\khehsutra-guru\badminton-highlight-indexer\backend\eval\ablation.py-462-
return experiment.load_golden_videos(
C:\Users\avidu\Projects\khehsutra-guru\badminton-highlight-indexer\backend\eval\ablation.py-463-
features_dir,
--
C:\Users\avidu\Projects\khehsutra-guru\badminton-highlight-indexer\backend\eval\ablation.py-467-

```



```

audio_policy="tolerate-missing",
C:\Users\avidu\Projects\khehsutra-guru\badminton-highlight-indexer\backend\eval\ablation.py-468-
skip_logger=logger,
backend\eval\ablation.py:469:         skip_message="ablation.load_videos: skipping %s ?
windowing-only schema "
C:\Users\avidu\Projects\khehsutra-guru\badminton-highlight-indexer\backend\eval\ablation.py-470-
"(no shuttle/person cue dicts), not an ablatable fusion-feature file",
C:\Users\avidu\Projects\khehsutra-guru\badminton-highlight-indexer\backend\eval\ablation.py-471-
)
--
C:\Users\avidu\Projects\khehsutra-guru\badminton-highlight-
indexer\backend\eval\annotations.py-47-   Delegates to the canonical loader (gt_loader.py,
ADR-008 prereq), which accepts BOTH
backend\eval\annotations.py-48-   golden conventions: positional/headered ``start,end`` (the
collector export) and the
backend\eval\annotations.py:49:   rally-annotator ``rally_number,start_time,end_time,...``
schema. Strict: raises
C:\Users\avidu\Projects\khehsutra-guru\badminton-highlight-
indexer\backend\eval\annotations.py-50-   ValueError on malformed rows (bad timestamp, start >=
end) so labeling mistakes
C:\Users\avidu\Projects\khehsutra-guru\badminton-highlight-
indexer\backend\eval\annotations.py-51-   surface immediately rather than silently skewing
scores. Returns intervals sorted by
--
backend\eval\classifier.py-5-installed. This implements just enough: standardized features,
balanced-class
C:\Users\avidu\Projects\khehsutra-guru\badminton-highlight-
indexer\backend\eval\classifier.py-6-gradient descent, sigmoid probabilities, and JSON
(de)serialization so a trained
backend\eval\classifier.py:7:model is a small, human-readable, version-controllable file.
C:\Users\avidu\Projects\khehsutra-guru\badminton-highlight-indexer\backend\eval\classifier.py-8-

```

[Showing results with pagination = limit: 30]

25. assistant (2026-07-04T00:45:53.343Z)

Perfect. Now let me understand the full angle here. Let me check the docs for open question 5:

26. user (2026-07-04T00:45:54.255Z)

220:5. **Engineering:** Should A9 close on the current Gen-0 non-serving bundle, or should the actual TemporalMaxer scorer swap be implemented before marking A9 done?

27. assistant (2026-07-04T00:45:59.583Z)

Now let me understand what question 5 was originally. Let me search for older versions:

28. user (2026-07-04T00:46:00.600Z)

9. Open questions

1. **Owner:** Is the first alpha allowed to use manual/offline corpus promotion, or should P10/P11 be reopened before any tester wave?

29. user (2026-07-04T00:46:03.425Z)

A4 nightly baseline refresh record:

```

```powershell
python scripts\nightly_regression.py --manifest output\human_lovo_manifest.json --features-dir
output --update-baseline
```

```

Result:

```

```text
INFO baseline snapshot written to C:\Users\avidu\Projects\khelsutra-guru\badminton-highlight-
indexer\eval_baselines\nightly_baseline.json
INFO default: n=15 tolF1=0.364 f1@0.5=0.472 merge_rate=0.135
INFO milestone: n=15 tolF1=0.364 f1@0.5=0.472 merge_rate=0.135
INFO baseline_off: n=15 tolF1=0.138 f1@0.5=0.237 merge_rate=0.638
```

```

Pass-mode proof:

```

```powershell
python scripts\nightly_regression.py --manifest output\human_lovo_manifest.json --features-dir
output --no-report
```

```

Result: `PASS`, exit code `0`, baseline snapshot `2026-07-03`, default/milestone `tol\_f1 = 0.3636`, `f1@0.5 = 0.4723`, `merge\_rate = 0.1350`; baseline-off `tol\_f1 = 0.1382`, `f1@0.5 = 0.2366`, `merge\_rate = 0.6380`.

A5 heuristic floor refresh record:

```

```powershell
python -m backend.eval.calibrate_local --manifest output\human_lovo_manifest.json --metric f1
--floor-out eval_baselines\heuristic_lovo_n15.json
```

```

Result:

```

```text
baseline (untuned) mean held-out f1: 0.237
calibrated mean held-out f1: 0.446 +/- 0.239 (gap +0.000)
```

```

The committed floor is `eval\_baselines/heuristic\_lovo\_n15.json`: `n = 15`, `mean\_f1 = 0.4456`, `std = 0.2390`, `metric = f1`, `iou = 0.5`. Every leave-one-video-out fold selected `(0.05, 1.0, 2.0, 0)`.

A6 ablation/R2 record:

```

```powershell
python -m backend.eval.ablation --features-dir output --granularity group
python -m backend.eval.rally_seg_eval --manifest output\human_lovo_manifest.json --features-dir
output --preset served --max-window-duration 0 --inactivity-end 0 --vs served --vs-max-window-
duration 6 --vs-inactivity-end 1.5 --json-out output\a6_r2_ablation.json
```

```

Result: the ablation CLI now runs, but full-fusion feature ablation still reports `n=6` because only the original six feature JSONs carry the full fusion schema; A7 must regenerate/upconvert the newer nine before fusion/person/audio promotion decisions. The n=15 R2/tolF1 contrast is positive for the served milestone versus baseline-off: baseline/off `tolF1 = 0.138`, served `tolF1 = 0.364`, `Delta tolF1 = +0.225` with CI `[+0.165, +0.283]`, sign `14/0`, `p = 0.000`; `Delta F1@0.5 = +0.236` with CI `[+0.161, +0.313]`; merge rate improves from `0.638` to `0.135`. Net over-segmentation guard passes (`1.276 -> 0.378`), while the strict segment-count rule remains blocked by increased splits, so R2/tolF1 is the supported gate for the baseline-off to served milestone contrast rather than a blanket fusion/default-flip approval.

9. Open questions

30. user (2026-07-04T00:46:06.525Z)

9. Open questions

1. **Owner:** Is the first alpha allowed to use manual/offline corpus promotion, or should P10/P11 be reopened before any tester wave?

2. ****Owner/platform:**** Which hosted compute target is the next stand-up: Cloud Run GPU, GCE L4 VM, or another approved GPU host?
3. ****Owner/legal:**** What is the acceptable alpha wording while WASB C3 remains unresolved?
4. ****Engineering:**** Should the portable corpus manifest live as a generated artifact, a checked-in redacted manifest, or both?
5. ****Engineering:**** Should n=15 feature regeneration target full fusion schema for every clip, or should the eval loaders support mixed schemas with explicit capability tags?

10. Definition of done

- [] Invited tester can complete the hosted record/upload -> reel download CUJ without SSH.

31. assistant (2026-07-04T00:46:23.376Z)

Perfect! Now I understand all four angles. Let me now check for issues. Let me examine the specific altitude issues:

****Angle 1:**** Is question 5 about schema tags being addressed or papered over?

- The `is_full_fusion_feature_file()` does runtime schema-sniffing by checking if candidates have "shuttle" and "person" keys
- This detects the schema at runtime by re-reading the JSON instead of explicit versioning written at generation time
- The question asked whether loaders should support "mixed schemas with explicit capability tags"
- The PR does NOT add explicit tags; it adds better runtime detection

****Angle 2:**** Is `attach_source_video_paths` a second path source of truth?

- `golden_manifest.py` resolves paths at load time using trajectory/labels basename matching
- `attach_source_video_paths` adds a SECOND path provider from the collector source manifest
- Both mechanisms independently provide `video_path`, creating layering

****Angle 3:**** Is thermal safety in the right layer?

- `max_gpu_temp_c` is implemented in `extract_video`'s `_progress` callback via telemetry
- This lives in eval-tool specific code, not in shared detector/telemetry
- Only `fusion_golden` has this, not other GPU passes

****Angle 4:**** Does marking A9 IN PROGRESS misrepresent status?

- A9 says "Run Gen-0/TemporalMaxer shadow training"
- The PR marks A9 as IN PROGRESS but TemporalMaxer is not implemented
- The note says "TemporalMaxer itself is not implemented in this branch"
- The status says it's IN PROGRESS but the core deliverable scope question is punted

Now let me create the final analysis as JSON:

```
```json
[
 {
 "file": "backend/eval/fusion_golden.py",
 "line": 42,
 "summary": "is_full_fusion_feature_file() does runtime schema-sniffing instead of explicit version tags written at generation time. Docs §9 Q5 asked exactly this: should eval loaders support mixed schemas with explicit capability tags? This PR detects capability by re-reading JSON structure rather than stamping explicit schema version at write time.",
 "failure_scenario": "Future tooling cannot distinguish full-fusion vs windowing-only files without opening and parsing JSON; if feature JSONs ever migrate formats, schema detection becomes fragile. Consuming code must always assume re-reading may fail or that two files with same name pattern may have incompatible structures."
 },
 {
 "file": "backend/eval/fusion_golden.py",
 "line": 263,
 "summary": "attach_source_video_paths() introduces a SECOND path resolution layer on top of golden_manifest.py's resolver. Golden_manifest owns trajectory/labels path resolution via basename matching; attach_source_video_paths adds source_manifest as authoritative for"
 }
]
```

```

video_path. This stacks two path-attach mechanisms with no explicit priority or conflict-
resolution rule.",
 "failure_scenario": "If golden_manifest resolver and source_manifest provide different
video_path values for the same spec, spec['video_path'] is silently overwritten. Debuggers
cannot tell whether a path came from the manifest resolver or collector manifest. A third path
source added later becomes even harder to reason about.",
},
{
 "file": "backend/eval/fusion_golden.py",
 "line": 200,
 "summary": "max_gpu_temp_c thermal abort is wired through extract_video's _progress
callback, which is eval-tool-specific code. Thermal safety should live in the shared detector or
telemetry layer so all GPU passes get protection, not just fusion_golden. This leaves
fusion_compare, fusion_audio, and other GPU consumers unprotected if they call PersonDetector
directly.",
 "failure_scenario": "A long-running fusion_audio augmentation or another GPU tool that never
checks thermal telemetry can overheat hardware or trigger OS thermal shutdown with no graceful
abort. Repeated thermal kills mask the real problem if only some eval tools guard temperature."
},
{
 "file": "docs/ALPHA_LAUNCH_READINESS.md",
 "line": 109,
 "summary": "A9 is marked 'IN PROGRESS | this PR' but A9's scope definition is 'Run
Gen-0/TemporalMaxer shadow training' and the PR explicitly notes 'TemporalMaxer itself is not
implemented in this branch' (line 212). The tracker's S7 conventions say status reflects actual
deliverable completion. Marking A9 IN PROGRESS while the named scorer-swap feature remains
unimplemented misrepresents progress per the tracker's own definition.",
 "failure_scenario": "Downstream decision-makers see 'A9 IN PROGRESS' and assume
TemporalMaxer is being actively developed in this PR. They don't see the core scope item is
deferred to a follow-up. The docs then say 'evidence supports either closing A9 as a Gen-0
shadow baseline or opening a follow-up M0.4 scorer-swap PR' ? but the tracker claims A9 is
already IN PROGRESS, creating conflicting signals about whether the real work is scoped here or
punted."
}
]
...

```

Wait, I should verify line numbers more carefully and check if there are any other issues. Let me look at the tracker table more carefully:

## 32. user (2026-07-04T00:46:24.396Z)

@@ -10,7 +10,7 @@

## 0. TL;DR

-The corpus is now large enough to move from "grow labels first" into alpha-readiness work, but the 15-video headline is not itself a launch gate pass. ``[verified]`` The golden manifest has 15 badminton videos; ``[analysis]`` the local manifest needed path rebasing before manifest-driven commands ran; ``[analysis]`` only 6 of 15 feature JSONs had the full fusion schema; ``[analysis]`` the n=15 quality baseline is stale and must be intentionally refreshed before it can gate anything.

+The corpus is now large enough to move from "grow labels first" into alpha-readiness work, but the 15-video headline is not itself a launch gate pass. ``[verified]`` The golden manifest has 15 badminton videos; ``[analysis]`` A2-A7 made the local manifest/path and full-fusion/audio feature surface reproducible on all 15; ``[analysis]`` A4/A5 refreshed the n=15 nightly and heuristic floors; ``[analysis]`` A8/A9 still reject served Gate A default promotion and owned-model serving claims on the current evidence.

The first controlled alpha should be a hosted, authenticated product loop: upload a video, process it, select rallies, compile, and download a reel. The owned-model/commercial rally-detection claim remains a separate NO-GO until the ship-gate target is ratified and the WASB checkpoint provenance is resolved.

@@ -31,8 +31,9 @@ This doc separates those into executable workstreams. The goal is to reach an

```

in
| D1 | **Product alpha and owned-model launch are separate decisions.** | `NEXT_STEPS.md` NO-GO
banner, 2026-06-30; verified 2026-07-03 | We can launch a controlled hosted product loop without
claiming the owned model is commercially cleared. |
--
- `[analysis]` The local manifest had stale absolute paths. A temporary rebased manifest found
all 15 trajectories and labels in this workspace.
-- `[analysis]` There are 15 `output/*_golden_features.json` files, but only 6 carry full
shuttle/person/audio candidate features. The newer 9 are windowing-only schema and are skipped
by fusion/person eval.
-- `[analysis]` `python -m backend.eval.fusion_compare --features-dir output` loaded only 6
videos and measured person-fusion `Delta F1 = -0.021`; not promotable.
-- `[analysis]` `python -m backend.eval.serve_contrast --manifest <rebased-n15>` measured
proposal Gate A mean `Delta F1 +0.0287`, but served mean `Delta F1 -0.0328`; not served-safe.
-- `[analysis]` `python -m training.gen0.harness --manifest <rebased-n15> --floor
eval_baselines/heuristic_lovo_n6.json --version analysis-n15 --out <temp-out>` produced learned
LOVO mean F1 `0.551`, `ship=False`; this pre-A5 run used the stale n=6 floor, so future
comparisons should use `eval_baselines/heuristic_lovo_n15.json`.
+- `[analysis]` A7 generated full shuttle/person fusion schema for all 15 videos and A8 audio
augmentation populated audio features for all 15.
+- `[analysis]` `python -m backend.eval.fusion_compare --features-dir output` now runs over all
15 and measures person-fusion `Delta F1 = +0.008`; the paired CI crosses zero, so person fusion
is not promotable by itself.
+- `[analysis]` `python -m backend.eval.fusion_audio --features-dir output --compare` measures
audio `Delta F1 = +0.018`; the paired CI crosses zero, so audio is a shadow/suggestive cue, not
a default-flip.
+- `[analysis]` `python -m backend.eval.serve_contrast --manifest
output\human_lovo_manifest.json` measured proposal Gate A mean `Delta F1 +0.0287`, but served
mean `Delta F1 -0.0328`; not served-safe.
+- `[analysis]` `python -m training.gen0.harness --manifest output\human_lovo_manifest.json
--floor eval_baselines\heuristic_lovo_n15.json --version 0.1.1-n15-shadow.20260704 --out
artifacts\rally_tal_gen0` produced learned LOVO mean F1 `0.551`, floor passed versus heuristic
`0.446`, but sign test `9/15`, `p = 0.3036`, `ship=False`.
- `[analysis]` `python scripts/nightly_regression.py --manifest <rebased-n15> --features-dir
output --no-report` reported default/milestone tolF1 `0.3636` vs baseline-off tolF1 `0.1382`,
but marked the frozen baseline stale due to config/corpus changes.
- `[analysis]` Before A6, `python -m backend.eval.ablation --features-dir output --granularity
group` failed at import due to a circular import between `backend/eval/ablation.py` and
`backend/eval/ablation_pdf.py`; A6 fixes the module entrypoint and keeps full-fusion ablation
scoped to the 6 videos with full schema until A7.

@@ -97,13 +99,13 @@ Legend: TODO = not started, IN PROGRESS = actively being worked, DONE =
shipped/
| A0 | Create this alpha readiness tracker and index it in docs | `badminton-highlight-indexer`
| - | No | DONE | #451 |
| A1 | Stand up alpha serving endpoint: Cloud Run/GPU or approved host, health check, edge auth
ON, upload -> process -> jobs -> compile -> download verified | `khelsutra` + `badminton-
highlight-indexer` | A0 | Owner spend / CF config | TODO | - |
| A2 | Make the n=15 golden manifest portable/reproducible; add a path/corpus smoke check that
fails loudly when labels or trajectories are missing | `badminton-highlight-indexer` | A0 | No |
DONE | #452 |
-| A3 | Promote the 15-video source-video/MD5 records into the collector/vault path; reconcile
collector's six-video source manifest with the 15-video eval corpus | `sports-data-collector` +
vault | A2 | Owner upload/storage scope | TODO | - |
+| A3 | Promote the 15-video source-video/MD5 records into the collector/vault path; reconcile
collector's six-video source manifest with the 15-video eval corpus | `sports-data-collector` +
vault | A2 | Owner upload/storage scope | IN PROGRESS | collector #62 |
| A4 | Refresh the n=15 nightly regression baseline intentionally and record the exact
command/output; keep stale-baseline warnings until reviewed | `badminton-highlight-indexer` | A2
| Reviewer sign-off | DONE | #453 |
| A5 | Recompute the heuristic n=15 floor and replace the stale `heuristic_lovo_n6` floor for
future Gen-0 comparisons | `badminton-highlight-indexer` | A2 | No | DONE | #454 |
-| A6 | Fix `backend.eval.ablation` CLI circular import, then run the R2/tolF1 ablation on the
n=15 corpus | `badminton-highlight-indexer` | A2 | No | IN PROGRESS | this PR |
-| A7 | Regenerate or upconvert full-fusion feature JSONs for the newer 9 videos so

```

```

fusion/person/audio analyses run over all 15 | `badminton-highlight-indexer` | A2 | GPU/data
availability | TODO | - |
-| A8 | Re-run `fusion_compare`, group ablation, and served contrast after A7; explicitly
promote/reject person fusion, Gate A served-default, and any other R-series default flips |
`badminton-highlight-indexer` | A6, A7 | Reviewer sign-off | TODO | - |
-| A9 | Run Gen-0/TemporalMaxer shadow training against the refreshed n=15 floor; produce a non-
serving artifact bundle and gate verdict | `badminton-highlight-indexer` | A5, A8 | Owner
approval for M0.4 scope | TODO | - |
+| A6 | Fix `backend.eval.ablation` CLI circular import, then run the R2/tolF1 ablation on the
n=15 corpus | `badminton-highlight-indexer` | A2 | No | DONE | #455 |
+| A7 | Regenerate or upconvert full-fusion feature JSONs for the newer 9 videos so
fusion/person/audio analyses run over all 15 | `badminton-highlight-indexer` | A2 | GPU/data
availability | IN PROGRESS | this PR |
+| A8 | Re-run `fusion_compare`, group ablation, and served contrast after A7; explicitly
promote/reject person fusion, Gate A served-default, and any other R-series default flips |
`badminton-highlight-indexer` | A6, A7 | Reviewer sign-off | IN PROGRESS | this PR |
+| A9 | Run Gen-0/TemporalMaxer shadow training against the refreshed n=15 floor; produce a non-
serving artifact bundle and gate verdict | `badminton-highlight-indexer` | A5, A8 | Owner
approval for M0.4 scope | IN PROGRESS | this PR |
| A10 | Decide and record the alpha ship-gate target: metric, corpus slice, minimum threshold,
significance rule, and what claims are allowed | Owner + `badminton-highlight-indexer` | A4, A5,
A8 | Owner decision | GATED | - |
| A11 | Resolve or explicitly defer WASB C3 provenance for alpha messaging; if deferred,
document "no commercial model sharing" copy guardrail | Owner/legal | A10 | Legal/provenance |
GATED | - |
| A12 | Reopen or defer P10/P11 corpus flywheel: `PATCH /api/segments`, collector promotion,
`train`/`reeval` job, delta-F1 surface, and alignment with `COMPUTE_DECOUPLED_SERVING` M11 |
`khelsutra` + `badminton-highlight-indexer` + collector | A1, A3 | Owner D13 reopen
(`khelsutra/docs/STUDIO_MEDIA_LIFECYCLE.md`) | GATED | - |
@@ -174,22 +176,57 @@ python -m backend.eval.rally_seg_eval --manifest
output\human_lovo_manifest.json

```

Result: the ablation CLI now runs, but full-fusion feature ablation still reports `n=6` because only the original six feature JSONs carry the full fusion schema; A7 must regenerate/upconvert the newer nine before fusion/person/audio promotion decisions. The n=15 R2/tolF1 contrast is positive for the served milestone versus baseline-off: baseline/off `tolF1 = 0.138`, served `tolF1 = 0.364`, `Delta tolF1 = +0.225` with CI `[+0.165, +0.283]`, sign `14/0`, `p = 0.000`; `Delta F1@0.5 = +0.236` with CI `[+0.161, +0.313]`; merge rate improves from `0.638` to `0.135`. Net over-segmentation guard passes (`1.276 -> 0.378`), while the strict segment-count rule remains blocked by increased splits, so R2/tolF1 is the supported gate for the baseline-off to served milestone contrast rather than a blanket fusion/default-flip approval.

+A7 full-fusion/audio feature refresh record:

```

+
+```powershell
+python -m backend.eval.fusion_golden --manifest output\human_lovo_manifest.json --out-dir
output --source-manifest ..\sports-data-collector\data\golden_source_videos.json --missing-
fusion-only --smallest-first --max-gpu-temp-c 87
+python -m backend.eval.fusion_audio --manifest output\human_lovo_manifest.json --features-dir
output --source-manifest ..\sports-data-collector\data\golden_source_videos.json --augment
+```
+
+Result: the GPU pass completed without a thermal abort after fans were set to max performance.
Full-fusion schema is now present for `15/15` feature JSONs, and audio features are present for
`15/15`. The large regenerated clips completed as follows: `mahadevpura_1` `1` window / `0`
positives / `10` GT, `GX020094` `99` / `30` / `40`, `GX030094` `79` / `21` / `41`, `GX010141`
`26` / `11` / `29`, `GX010137` `56` / `31` / `34`, `Boxhill_Doubles` `59` / `12` / `43`,
`mahadevpura_2` `14` / `2` / `40`, `GX010128` `51` / `15` / `52`, `Badminton_BXH_2` `29` / `1` /
`44`. The source manifest supplied all 15 source-video paths, including the F-drive Boxhill
originals.

```

+A8 fusion/default-decision record:

```

+
+```powershell
+python -m backend.eval.fusion_compare --features-dir output

```

```

+python -m backend.eval.fusion_audio --features-dir output --compare
+python -m backend.eval.ablation --features-dir output --granularity group
--
+- Audio is **shadow/suggestive, not promoted**: shuttle+person F1 `0.311`, shuttle+person+audio
F1 `0.329`, `Delta F1 +0.018`, paired CI `[-0.017, +0.053]`, sign `10W/4L`, `p = 0.180`.
+- Group ablation remains non-significant for cue promotion: audio `+0.018` CI `[-0.017,
+0.053]`; shuttle `+0.013` CI `[-0.002, +0.029]`; gate_a `-0.001` CI `[-0.012, +0.008]`; person
`-0.009` CI `[-0.045, +0.021]`.
+- Gate A served default is **rejected**: proposal mean `Delta F1 +0.0287`, CI `[-0.0222,
+0.0680]`; served mean `Delta F1 -0.0328`, CI `[-0.0799, +0.0052]`, `regresses=True`.
+- R2/cap/inactivity remains the strong quality result: served candidate F1 `0.472` vs baseline-
off `0.237`, `Delta F1 +0.236`, CI `[+0.161, +0.313]`; tolF1 `0.364` vs `0.138`, `Delta tolF1
+0.225`, CI `[+0.165, +0.283]`; merge rate `0.135` vs `0.638`. Net over-seg guard passes, strict
per-video split-count guard still blocks blanket promotion language.
+
+A9 Gen-0 shadow-training record:
+
+```powershell
+python -m training.gen0.harness --manifest output\human_lovo_manifest.json --floor
eval_baselines\heuristic_lovo_n15.json --out artifacts\rally_tal_gen0 --version
0.1.1-n15-shadow.20260704
+```
+
+Result: current Gen-0 LogReg/TAL harness produced local bundle
`artifacts\rally_tal_gen0\0.1.1-n15-shadow.20260704` with learned LOVO mean F1 `0.551` versus
heuristic floor `0.4456`, so `floor_passed=True`. The gate remains **non-serving**: paired sign
test `9/15`, `p = 0.3036`, `ship=False`; blockers are uncalibrated ship-gate target #172 and
WASB checkpoint provenance C3. TemporalMaxer itself is not implemented in this branch; the
evidence supports either closing A9 as a Gen-0 shadow baseline or opening a follow-up M0.4
scorer-swap PR.
+
9. Open questions

1. **Owner:** Is the first alpha allowed to use manual/offline corpus promotion, or should
P10/P11 be reopened before any tester wave?
2. **Owner/platform:** Which hosted compute target is the next stand-up: Cloud Run GPU, GCE L4
VM, or another approved GPU host?
3. **Owner/legal:** What is the acceptable alpha wording while WASB C3 remains unresolved?
4. **Engineering:** Should the portable corpus manifest live as a generated artifact, a
checked-in redacted manifest, or both?
5. **Engineering:** Should n=15 feature regeneration target full fusion schema for every clip,
or should the eval loaders support mixed schemas with explicit capability tags?
+5. **Engineering:** Should A9 close on the current Gen-0 non-serving bundle, or should the
actual TemporalMaxer scorer swap be implemented before marking A9 done?

10. Definition of done

- [] Invited tester can complete the hosted record/upload -> reel download CUJ without SSH.
- [] Alpha service is gated behind the chosen auth posture and has a documented
disable/rollback path.
--
- [] P10/P11 are either reopened with a committed sequence or explicitly deferred for alpha-
plus.
@@ -226,3 +263,4 @@ python -m backend.eval.rally_seg_eval --manifest
output\human_lovo_manifest.json
Changelog

- 2026-07-03 - Created from a synced five-repo review of `NEXT_STEPS`, `ROADMAP`, golden-corpus
docs, local eval artifacts, and the current 15-video corpus state.
+- 2026-07-04 - Recorded A7 full-fusion/audio regeneration, A8 n15 fusion/default decisions, and
A9 Gen-0 n15 shadow-training gate verdict.

```

**33. assistant (2026-07-04T00:46:37.836Z)**

Perfect! Now I have all the details. Let me finalize my analysis:

```

```json
[
  {
    "file": "backend/eval/fusion_golden.py",
    "line": 42,
    "summary": "is_full_fusion_feature_file() detects full-fusion capability at runtime by re-reading JSON structure (checking for shuttle/person keys in candidates), rather than stamping explicit schema/version tags at generation time. Docs S9 Q5 (original) directly asked whether to support mixed schemas with explicit capability tags or runtime detection?this PR implements the runtime approach but leaves the question unresolved.",
    "failure_scenario": "Future consumers always need to open and parse JSON to detect capability; schema evolution becomes fragile if JSON structure can't reliably indicate schema version. If a consuming tool (fusion_audio, ablation) fails to call is_full_fusion_feature_file() before processing, it silently skips windowing-only files instead of failing loudly, hiding data-completeness gaps."
  },
  {
    "file": "backend/eval/fusion_golden.py",
    "line": 263,
    "summary": "attach_source_video_paths() introduces a second path-resolution layer stacked on top of golden_manifest.py's manifest resolver. The manifest resolver derives video_path from trajectory/labels filenames; attach_source_video_paths overwrites video_path with values from the collector source manifest. Two independent path sources with no conflict-resolution protocol.",
    "failure_scenario": "If golden_manifest resolver and source_manifest supply different video_path values for the same spec, the source_manifest silently wins without logging the override. Future consumers cannot tell whether video_path came from the golden manifest or the collector source. A third path provider added later becomes even harder to reason about without centralizing path resolution in golden_manifest.py."
  },
  {
    "file": "backend/eval/fusion_golden.py",
    "line": 205,
    "summary": "max_gpu_temp_c thermal abort is implemented in extract_video's _progress callback (eval-tool-specific code), protecting only fusion_golden extraction. Thermal safety should live in a shared detector or telemetry layer so all GPU-consuming eval tools (fusion_compare, fusion_audio, ablation on large videos) get the same protection.",
    "failure_scenario": "fusion_audio.augment() calls detect_hits_in_video() which may invoke GPU code without thermal monitoring. A long extraction run in a different tool or a manual detector call outside fusion_golden can overheat without graceful abort. Thermal shutdown is non-deterministic and hard to reproduce if only one eval tool guards temperature."
  },
  {
    "file": "docs/ALPHA_LAUNCH_READINESS.md",
    "line": 109,
    "summary": "A9 status is marked 'IN PROGRESS | this PR' but A9's deliverable is 'Run Gen-0/TemporalMaxer shadow training'. The PR's own text (line 212) explicitly states 'TemporalMaxer itself is not implemented in this branch' and asks whether to close A9 as Gen-0-only or open a follow-up PR for scorer-swap. Marking A9 IN PROGRESS while the named core feature is unimplemented violates the tracker's S7 convention that status reflects actual deliverable completion, not partial runs.",
    "failure_scenario": "Stakeholders see 'A9 IN PROGRESS' and assume TemporalMaxer implementation is underway. They miss that the scope question is deferred; the core named deliverable is punted to a future PR. A10/A11 gating on A9 completion creates a false dependency on a task whose main scope item is explicitly unimplemented."
  }
]
```

```

Actually, let me verify the exact line number for the TemporalMaxer mention:

#### 34. user (2026-07-04T00:46:39.131Z)

212:Result: current Gen-0 LogReg/TAL harness produced local bundle  
`artifacts\rally\_tal\_gen0\0.1.1-n15-shadow.20260704` with learned LOVO mean F1 `0.551` versus



heuristic floor `0.4456`, so `floor\_passed=True`. The gate remains **non-serving**: paired sign test `9/15`, `p = 0.3036`, `ship=False`; blockers are uncalibrated ship-gate target #172 and WASB checkpoint provenance C3. TemporalMaxer itself is not implemented in this branch; the evidence supports either closing A9 as a Gen-0 shadow baseline or opening a follow-up M0.4 scorer-swap PR.

### 35. assistant (2026-07-04T00:46:41.758Z)

Perfect. Now let me verify the critical line for question 5 being replaced:

### 36. user (2026-07-04T00:46:42.410Z)

5. **Engineering**: Should A9 close on the current Gen-0 non-serving bundle, or should the actual TemporalMaxer scorer swap be implemented before marking A9 done?

### 37. assistant (2026-07-04T00:46:56.468Z)

Perfect. Now let me compile the final JSON array with verified line numbers:

```
[
 {
 "file": "backend/eval/fusion_golden.py",
 "line": 42,
 "summary": "is_full_fusion_feature_file() detects full-fusion capability via runtime JSON schema-sniffing (checking for shuttle/person keys) rather than explicit schema/version tags at generation time. Docs §9 Q5 (original, pre-PR line 220 now) asked exactly this: should loaders support mixed schemas with explicit capability tags or runtime detection? This PR chooses runtime detection without adding explicit schema versioning at write time.",
 "failure_scenario": "Future consumers always need to re-read and parse JSON to detect schema capability; schema evolution becomes brittle if JSON structure alone must indicate version. If consuming code (fusion_audio, ablation) fails to call is_full_fusion_feature_file() upfront, it silently skips windowing-only files instead of failing loudly, hiding data gaps."
 },
 {
 "file": "backend/eval/fusion_golden.py",
 "line": 263,
 "summary": "attach_source_video_paths() stacks a second path-resolution layer on top of golden_manifest.py's resolver. The manifest resolver derives paths from trajectory/labels filenames; attach_source_video_paths overwrites video_path from collector source manifest. Two independent path sources for the same field with no explicit ownership or conflict-resolution rule.",
 "failure_scenario": "If golden_manifest and source_manifest supply different video_path values for the same spec, source_manifest silently wins. Consumers cannot tell whether video_path came from manifest resolution or collector source. Centralizing path resolution in golden_manifest.py is the deeper alternative: let it own video_path from all sources instead of having fusion_golden attach paths post-hoc."
 },
 {
 "file": "backend/eval/fusion_golden.py",
 "line": 205,
 "summary": "max_gpu_temp_c thermal abort lives in extract_video's _progress callback (eval-tool-specific code), protecting only fusion_golden. Thermal safety should live in a shared detector or telemetry layer so all GPU-consuming paths (fusion_compare, fusion_audio.detect_hits_in_video(), other detector calls) get uniform protection regardless of which eval tool invokes them.",
 "failure_scenario": "fusion_audio.augment() calls detect_hits_in_video() without thermal monitoring. A long GPU run in a different eval tool or a manual detector instantiation outside fusion_golden can overheat without graceful abort. Thermal shutdown is non-deterministic; only protecting one eval tool makes the failure hard to reproduce and leaves other tools vulnerable."
 },
 {
 "file": "docs/ALPHA_LAUNCH_READINESS.md",
 "line": 109,
 "summary": "A9 is marked 'IN PROGRESS | this PR' but A9's deliverable is 'Run
```

Gen-0/TemporalMaxer shadow training'. The PR documents 'TemporalMaxer itself is not implemented in this branch' (line 212) and Q5 (line 220) asks whether A9 should close as Gen-0-only or reopen for TemporalMaxer. Marking A9 IN PROGRESS while the named core feature is unimplemented violates tracker §7 convention that status reflects deliverable completion, not partial execution.",

"failure\_scenario": "Stakeholders see 'A9 IN PROGRESS' and assume TemporalMaxer implementation is actively being worked. They miss the scope question is unresolved and the core named feature is deferred. A10/A11 dependencies on A9 DONE create false critical-path coupling to a task whose core deliverable is explicitly punted to a follow-up PR."

```
}
]
```

<|end\_of\_solution|>

```json

```
[  
  {  
    "file": "backend/eval/fusion_golden.py",  
    "line": 42,  
    "summary": "is_full_fusion_feature_file() detects full-fusion capability via runtime JSON schema-sniffing (checking for shuttle/person keys) rather than explicit schema/version tags at generation time. Docs §9 Q5 (original, now line 220) asked exactly this: should loaders support mixed schemas with explicit capability tags or runtime detection? This PR chooses runtime detection without adding explicit schema versioning.",  
    "failure_scenario": "Future consumers always need to re-read and parse JSON to detect schema capability; schema evolution becomes brittle if JSON structure alone must indicate version. If consuming code (fusion_audio, ablation) fails to call is_full_fusion_feature_file() upfront, it silently skips windowing-only files instead of failing loudly, hiding data gaps."  
  },  
  {  
    "file": "backend/eval/fusion_golden.py",  
    "line": 263,  
    "summary": "attach_source_video_paths() stacks a second path-resolution layer on top of golden_manifest.py's resolver. The manifest resolver derives paths from trajectory/labels filenames; attach_source_video_paths overwrites video_path from collector source manifest. Two independent path sources for the same field with no explicit ownership or conflict-resolution rule.",  
    "failure_scenario": "If golden_manifest and source_manifest supply different video_path values, source_manifest silently wins. Consumers cannot tell whether video_path came from manifest resolution or collector source. Centralizing path resolution in golden_manifest.py is the deeper alternative: let it own video_path from all sources instead of having fusion_golden attach paths post-hoc."  
  },  
  {  
    "file": "backend/eval/fusion_golden.py",  
    "line": 205,  
    "summary": "max_gpu_temp_c thermal abort lives in extract_video's _progress callback (eval-tool-specific code), protecting only fusion_golden. Thermal safety should live in a shared detector or telemetry layer so all GPU-consuming paths (fusion_compare, fusion_audio.detect_hits_in_video(), other detector calls) get uniform protection regardless of which eval tool invokes them.",  
    "failure_scenario": "fusion_audio.augment() calls detect_hits_in_video() without thermal monitoring. A long GPU run in a different eval tool can overheat without graceful abort. Thermal shutdown is non-deterministic; only protecting one eval tool makes failure hard to reproduce and leaves other tools vulnerable."  
  },  
  {  
    "file": "docs/ALPHA_LAUNCH_READINESS.md",  
    "line": 109,  
    "summary": "A9 is marked 'IN PROGRESS | this PR' but A9 deliverable is 'Run Gen-0/TemporalMaxer shadow training'. The PR documents 'TemporalMaxer itself is not implemented in this branch' (line 212) and Q5 (line 220) asks whether A9 should close as Gen-0-only or reopen for TemporalMaxer. Marking A9 IN PROGRESS while the named core feature is unimplemented violates tracker §7 convention that status reflects deliverable completion.",  
    "failure_scenario": "Stakeholders see 'A9 IN PROGRESS' and assume TemporalMaxer
```

implementation is underway. They miss the scope question is unresolved and the core named feature is deferred to a follow-up PR. A10/A11 dependencies on A9 DONE create false critical-path coupling to a task whose main deliverable is explicitly punted."

```
    }  
]  
...
```